

Spectral factor models

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This draft: August 13, 2020

Abstract

We represent risk factors as sums of orthogonal components capturing fluctuations with cycles of different length. The representation leads to novel *spectral factor models* in which systematic risk is allowed (without being forced) to vary across frequencies. Frequency-specific systematic risk is captured by a notion of *spectral beta*. We show that traditional factor models restrict the spectral betas to be constant across frequencies. The restriction can hide horizon-specific pricing effects which spectral factor models are designed to reveal. We illustrate how the methods may lead to economically-meaningful dimensionality reduction in the factor space.

JEL classification: C22, C32, E32, G11, G12.

Keywords: systematic risk, factor models, frequency, cross-sectional asset pricing.

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We are grateful to the Editor, Bill Schwert, and an anonymous reviewer for very useful comments. We thank the participants in the Chicago Booth/Edhec conference on “New Methods for the Cross Section of Returns,” Chicago, September 28 2018, the conference “High Voltage Econometrics,” Palermo, October 4-5 2018, the conference “A Celebration of Peter Phillips’ 40 Years at Yale,” New Haven, October 19-20 2018, the Stevanovich Center conference “Market Microstructure and High-Frequency Data,” Chicago, May 2-4 2019, and the 12th Society for Financial Econometrics annual conference, Shanghai, June 11-14 2019. We also thank the discussant, Stefano Giglio, for his many suggestions as well as Andreas Neuhierl and Sydney Ludvigson for helpful communications.

1. Introduction

Systematic risk changes across investment styles but may also change across frequencies. While most of the finance literature has focused on the relation between risk and style, several stimulating contributions have offered evidence of the link between risk and frequency using alternative methods of inference and in different contexts.¹ Against this backdrop, a modeling and inferential framework which generalizes existing factor models by explicitly formalizing the interplay between risk, style and frequency is of economic value. The provision of such a framework is the subject of this paper.

Our approach begins with the following observation: any covariance-stationary factor, i.e., any factor for which a classical Wold representation applies (Wold, 1938), always admits an equivalent orthogonal representation in which the factor can be modeled as an infinite sum of orthogonal frequency-specific components. Importantly, each component captures fluctuations of the original factor with a specific periodicity.

We employ the orthogonal representation of the factor to define a novel *spectral factor model* in which excess asset returns are directly linked to the orthogonal components of the original factor, rather than to the aggregate factor. In the model, systematic risk is allowed to change across different frequencies but, importantly, it is never forced to do so. More explicitly, we show that a traditional factor model may readily be viewed as a *restricted version* of our spectral factor model in which, as one would expect, the restriction is one of equal risk across frequencies.

We define *spectral beta* as the partial effect of individual orthogonal components of the factor on excess asset returns. Exploiting the orthogonal Wold representation, we show that the traditional beta between excess asset returns and the aggregate factor is a weighted average of spectral betas in which the weights depend on the informational content (i.e., the “relative” variance) of the factor’s Wold components.

Because of the orthogonality of the components, no cross-beta terms appear in the representation of the overall beta, thereby guaranteeing that all frequency-specific information is captured by the corresponding spectral beta. Furthermore, the orthogonality of the components implies that the betas obtained from a multiple regression of asset returns on the factor’s Wold components are the same as simple regression betas.² Thus, our setting seems

¹See, e.g., Hawawini (1983), Handa, Kothari, and Wasley (1989), Kothari, Shanken, and Sloan (1995), Daniel and Marshall (1997), Berkowitz (2001), Cogley (2001), Jagannathan and Wang (2007), Cohen, Polk, and Vuolteenaho (2009), Gilbert, Hrdlicka, Kalodimos, and Siegel (2014), Dew-Becker and Giglio (2016), Kamara, Korajczyk, Lou, and Sadka (2016), Kang, In, and Kim (2017), Brennan and Zhang (2018), Lettau, Ludvigson, and Ma (2018), Chaudhuri and Lo (2019), Neuhierl and Varneskov (2019) and Schneider (2019).

²Kan, Robotti, and Shanken (2013) formalize how the incremental (pricing) information in a factor cannot be deduced from multiple regression betas but, rather, from univariate regression betas. See, also, Cochrane

ideally suited to study whether a particular frequency-specific component makes an incremental contribution to the model's overall explanatory power, given the other components.

We study the estimation of the spectral components and spectral betas using methods which explicitly rely on the identification of the Wold components of the original covariance-stationary multivariate process.

Economically, our proposed spectral factor models may be viewed as adding a dimension, i.e., frequency, to style. They do so by relaxing the restriction of equal risk across frequency which is implicit in the way in which traditional factor models are expressed. Dispensing with the restriction may lead to dimensionality reduction in the factor space.

To this extent, we show that a business-cycle component of the market excess return process, i.e., a component with cycles spanning the 32 to 64-month horizon, is effective in pricing cross sections of test assets widely employed in the recent asset pricing literature, from the 25 Fama-French portfolios formed on size and book-to-market to 48 portfolios sorted on a broad set of 24 characteristics summarizing heterogeneity in expected returns.³ The corresponding price of risk is stable over alternative cross sections and statistically significant even when employing standard errors robust to model misspecification as in the work of Kan et al. (2013). Importantly, when comparing a spectral factor model in which the only factor is a business-cycle market component to the Fama and French (2015) five-factor model or to the four-factor model of Hou, Xue, and Zhang (2015), misspecification-robust tests of equal $\mathbb{R}^2$ generally lead to failure to reject the null (of equal $\mathbb{R}^2$ s) at all conventional levels. We argue that attention to frequency, in addition to style, may lead to parsimonious, as well as economically-sensible, factor structures.

What leads to frequency-specific risk, or horizon effects, in cross-sectional pricing? In recent work, using aggregation as a means to assess risk over alternative horizons/frequencies, Kamara et al. (2016) emphasize the importance of delayed price adjustments to the information in the systematic factors. Differently put, they emphasize the role played by lagged dependencies between excess asset returns and the factors, with lagged factors affecting excess asset returns.⁴

We study formally the link between a model in which systematic risk has a frequency

(2005, Chapter 13.4) for a related discussion using stochastic discount factors.

³We study a set of anomalies similar to that used in Hou, Xue, and Zhang (2020) and Kozak, Nagel, and Santosh (2020).

⁴There is additional evidence that information may not be incorporated into prices immediately. Hou and Moskowitz (2005) show that firms that respond to market returns with a delay command a significant return premium. Gilbert et al. (2014) find that the delayed revelation of the impact of systematic news on opaque firms constitutes a source of uncertainty which is priced at high frequencies. Hong, Torous, and Valkanov (2007) and Cohen and Frazzini (2008) point to investors' limited attention as a potential cause of lead-lag patterns in returns.

dimension, as in our framework, and a model in which returns slowly adjust to factor-specific information, as in the work of Kamara et al. (2016). We show that delayed price adjustments will, in general, lead to spectral betas changing across frequencies. Conversely, spectral factor models imply (forms of) delayed price adjustments.

In the data, we find that a delayed adjustment model with a one-month delay can yield, as an approximation, the spectral betas found across broadly-studied portfolios, such as the size and value-sorted portfolios. Long delays are, however, needed, in particular when focusing on value, in order to transition from an approximation to accurate assessments of *priced* frequency-specific systematic risk. We conclude with two observations. First, the literature on delayed adjustments has focused on short-term effects (see, e.g., the work of Hou and Moskowitz, 2005). We emphasize the role played by longer adjustments in leading to low-frequency dynamics important for pricing. Second, delayed adjustment models are statements about the return data generating process. Irrespective of the length of the adjustment and the likely-different economic logic associated with different lengths, spectral factor models may be viewed as a way to operationalize delayed adjustment models and translate them into the corresponding frequency-specific quantities of risk.

We illustrate our approach in Section 2. In spite of its simplicity, the illustration provides all necessary ingredients and intuition. We expand on positioning in Section 3. The rest of the paper is devoted to the substantive core of our treatment.

2. Spectral factor models: logic and intuition

Consider two covariance-stationary time series y_t (e.g., an excess asset return) and x_t (e.g., a factor).⁵ A traditional linear factor model is simply

$$y_t = \alpha + \beta x_t + u_t, \tag{1}$$

where u_t is an error term with standard properties. The classical beta is easily obtained from the linear projection of y_t onto a constant and x_t : $\beta = \frac{\mathbb{C}[y_t, x_t]}{\mathbb{V}[x_t]}$.

Assume, now, that both the excess asset return and the factor can be broken down into two *orthogonal* components capturing economic cycles shorter than 2^{j^*} periods (for example, months) and longer than 2^{j^*} periods, for some $j^* \geq 1$, respectively.⁶ In symbols,

⁵The factor may, of course, be multivariate. We work with a univariate factor for conciseness.

⁶We will later generalize to an unrestricted number of components.

$y_t = y_t^{<2^{j^*}} + y_t^{>2^{j^*}}$ and $x_t = x_t^{<2^{j^*}} + x_t^{>2^{j^*}}$ with

$$\mathbb{C}[y_t^{<2^{j^*}}, y_t^{>2^{j^*}}] = 0 = \mathbb{C}[x_t^{<2^{j^*}}, x_t^{>2^{j^*}}], \quad (2)$$

$$\mathbb{C}[y_t^{<2^{j^*}}, x_t^{>2^{j^*}}] = 0 = \mathbb{C}[x_t^{<2^{j^*}}, y_t^{>2^{j^*}}]. \quad (3)$$

Hence, the components are orthogonal for each process and across processes.⁷ From now on, we will dub the first component “high-frequency” or “HF” and the second component “low-frequency” or “LF”. We will, therefore, write x_t^{HF} in place of $x_t^{<2^{j^*}}$ and x_t^{LF} in place of $x_t^{>2^{j^*}}$.

In Section 4, we show that it is always possible to decompose covariance-stationary time series (like y_t and x_t) in such a way that Eq. (2) and Eq. (3) are satisfied. Our proposed decomposition has additional helpful features. First, it is adapted to time t information and is, therefore, non anticipative, a characteristic which makes it especially suitable for real-time out-of-sample applications. Second, contrary to analogous orthogonal representations using discrete Fourier transforms, the decomposition will be shown to be a general property of the data-generating process (with implications in terms of applicability and interpretability) rather than a property of the *transformed* data. In Section 5, we discuss how the decomposition can be operationalized.

Next, we posit the following regression model:

$$y_t = \alpha + \beta^{HF} x_t^{HF} + \beta^{LF} x_t^{LF} + u_t. \quad (4)$$

Eq. (4) is what we call a *spectral factor model*, a specification in which the factor is decomposed into orthogonal components capturing cycles of different length and, importantly, one in which the components of x_t have different partial effects. Because the components are orthogonal to each other (see Eq. (2)), these partial effects can be expressed as

$$\beta^{HF} = \frac{\mathbb{C}[y_t, x_t^{HF}]}{\mathbb{V}[x_t^{HF}]}, \quad (5)$$

$$\beta^{LF} = \frac{\mathbb{C}[y_t, x_t^{LF}]}{\mathbb{V}[x_t^{LF}]}. \quad (6)$$

In light of Eq. (3), they can also be expressed as

$$\beta^{HF} = \frac{\mathbb{C}[y_t^{HF}, x_t^{HF}]}{\mathbb{V}[x_t^{HF}]}, \quad (7)$$

$$\beta^{LF} = \frac{\mathbb{C}[y_t^{LF}, x_t^{LF}]}{\mathbb{V}[x_t^{LF}]}. \quad (8)$$

⁷The dyadic nature of their fluctuations will be justified in Section 4.

In other words, orthogonality permits us to compute the partial effects in two equivalent ways: (i) by projecting y_t on the frequency-specific components of x_t (as implied by Eq. (5) and Eq. (6)) or (ii) by projecting the frequency-specific components of y_t on the corresponding frequency-specific components of x_t (as implied by Eq. (7) and Eq. (8)). The latter is, of course, the result of regressions defined directly on the individual orthogonal components:

$$y_t^{HF} = \alpha + \beta^{HF} x_t^{HF} + u_t, \quad (9)$$

$$y_t^{LF} = \alpha + \beta^{LF} x_t^{LF} + u_t. \quad (10)$$

Since the components capture cycles of specific lengths, the betas in Eq. (7) and Eq. (8) may be interpreted as measures of systematic risk over the corresponding cycles. Eq. (7) and Eq. (8) define our notions of *spectral betas*.

The orthogonality of the components has a key implication in asset pricing. It is known that the incremental information of a factor in multi-factor asset pricing models cannot be measured by evaluating the statistical significance of the (second-stage) coefficients associated with multiple regression betas. Instead, one should be evaluating the statistical significance of the (second-stage) coefficients associated with regressions of expected returns on *univariate* regression betas, i.e., the prices of covariance risk (c.f., Kan et al., 2013, and the references therein). In spectral factor models, the multiple regression betas coincide with the univariate regression betas. Once these estimates are fed into a second-stage pricing equation, the corresponding risk prices can, therefore, be immediately interpreted to assess the relative contribution (to pricing) of individual (orthogonal) components.

Finally, the traditional beta is necessarily a weighted average of the spectral betas in Eq. (7) and Eq. (8) above. In fact,

$$\begin{aligned} \beta &= \frac{\mathbb{C}[y_t, x_t]}{\mathbb{V}[x_t]} \\ &= \frac{\mathbb{C}[y_t^{HF} + y_t^{LF}, x_t^{HF} + x_t^{LF}]}{\mathbb{V}[x_t]} \\ &= \frac{\mathbb{C}[y_t^{HF}, x_t^{HF}]}{\mathbb{V}[x_t]} + \frac{\mathbb{C}[y_t^{LF}, x_t^{LF}]}{\mathbb{V}[x_t]} \\ &= \frac{\mathbb{V}[x_t^{HF}]}{\mathbb{V}[x_t]} \frac{\mathbb{C}[y_t^{HF}, x_t^{HF}]}{\mathbb{V}[x_t^{HF}]} + \frac{\mathbb{V}[x_t^{LF}]}{\mathbb{V}[x_t]} \frac{\mathbb{C}[y_t^{LF}, x_t^{LF}]}{\mathbb{V}[x_t^{LF}]} \\ &= v^{HF} \beta^{HF} + v^{LF} \beta^{LF}, \end{aligned} \quad (11)$$

where the information ratios v^{HF} and v^{LF} can be interpreted as weights. Note, in fact, that $v^{HF} + v^{LF} = 1$.

The beta decomposition in Eq. (11) clarifies why the traditional beta may lead to misleading risk assessments. In essence, if frequency plays a role, risk is captured by the spectral betas. However, the information ratios matter in determining the traditional beta. Consider a fast-moving factor for which $v^{HF} \gg v^{LF}$. The corresponding (high-frequency) spectral beta will be highly weighed in the overall beta. The overall beta may, therefore, hide the signal (i.e., the risk) associated with the slow-moving component(s). Such signal, albeit small in volatility terms, may be important for asset pricing. This observation formalizes, using our methods, a central insight of the long-run risk literature (Bansal and Yaron, 2004).

We conclude this section by stressing that these simple derivations highlight a central aspect of our proposed approach. We capture frequency-specific systematic risk by suitably generalizing a traditional factor model. Our specification in Eq. (4) is an unrestricted version of Eq. (1). Should, in fact, risk be constant across frequencies, i.e., if $\beta^{HF} = \beta^{LF} = \beta$, then Eq. (1) would coincide with Eq. (4). Empirical investigations which begin with Eq. (4), rather than with Eq. (1), are therefore designed to let the data speak about the differential role played by alternative frequencies/horizons. Zooming onto individual spectral betas allows us to uncover signal in the assessment of risk (if any signal exists) and separate it from noise.

A complementary way to interpret the proposed framework is the following: if Eq. (1) represents a “single” or “univariate” factor model, Eq. (4) represents a “multi-factor” counterpart in which there is *no proliferation* in the number of *economic* factors. There is, however, explicit emphasis on frequency as a dimension of risk. Naturally, one could have a multi-factor model to begin with (in Eq. (1)) and perform a frequency-based decomposition of a multi-factor model (in Eq. (4)). Our intuition in this section, and the treatment in the following sections, would be unaltered.

Our emphasis in this section was on risk. When transitioning from risk to cross-sectional pricing, given Eq. (11), a spectral factor model with two components will have the same pricing implications as a traditional model only if $\lambda v^{HF} = \lambda^{HF}$ and $\lambda v^{LF} = \lambda^{LF}$, where λ is the price of risk on the aggregate beta in the standard model and λ^{HF} and λ^{LF} are the prices of risk on the high and low-frequency components in the spectral model. The same logic applies to multiple (i.e., more than two) components and multiple (i.e., more than one) factors.

3. Literature review

The observation that betas may change with frequency is not new in the literature. It has been empirically validated using a variety of inferential methods. It is, therefore, helpful to place our work in the context of four streams of the asset pricing literature.

The first set of contributions emphasizes that betas may change with the rebalancing frequency of the anomaly portfolio used as a test asset (see, e.g., Handa et al., 1989; Kothari et al., 1995; Cohen et al., 2009). As an example, Handa et al. (1989) find that the market-model betas computed using annually-rebalanced portfolio returns explain the cross section of size portfolios.

The second stream of papers uses (*without* changing the original rebalancing frequency) temporal aggregation (i.e., compounding) of the portfolio returns and the factor(s) of interest to study horizon effects in asset valuation (c.f., e.g., Hawawini, 1983; Daniel and Marshall, 1997; Jagannathan and Wang, 2007; Gilbert et al., 2014; Kamara et al., 2016; Lettau et al., 2018).

A third area of the literature exploits the informational potential of multi-resolution filters⁸ as a way to decompose returns and factors and obtain scale-specific (or, equivalently, frequency-specific) assessments of risk (e.g., Bandi and Tamoni, 2017, and Kang et al., 2017).

Finally, stimulating recent work identifies frequency-specific betas by employing methods of (band) spectral inference in the tradition of Hannan's early work (Hannan, 1963a, and Hannan, 1963b).⁹ Neuhierl and Varneskov (2019), in particular, show that the co-spectrum between returns and the stochastic discount factor features dependence on frequency through the transitory component of the stochastic discount factor.

While the empirical evaluation of frequency-specific betas will be central to our treatment, we view the paper's contribution as introducing a novel approach to factor models. The proposed factor structure is suited to specify and quantify frequency-specific risk by generalizing the broad class of covariance-stationary factor structures. Said differently, the paper's contribution is a *data generating process*, one which will be used to contribute to the existing body of work in cross-sectional asset pricing from an alternative vantage point.

Returning to the work on rebalancing frequency, we show that frequency-specific effects can be obtained, in our framework, without rebalancing.

The work on aggregation and multiresolution analysis has provided important guidance about the dimension along which existing factor structures may fail (see, e.g., the insights in Gilbert et al., 2014, and Kamara et al., 2016) but it has not resulted in an operational approach to the specification of factor models. Such a specification is central to applied work. It is, for instance, important in the context of asset allocation. We, instead, begin with a modeling framework which adds a frequency-dimension to existing factor structures and provide a direct (regression-based) method to extract frequency-specific signal over different

⁸An extensive introduction to multiresolution methods with an emphasis on economic and financial applications is contained in Gençay, Selçuk, and Whitcher (2001).

⁹For early applications of band spectral regression to economic problems, see Engle (1974).

frequencies in a cohesive manner. In this sense, the proposed modeling framework is complementary to the informative empirical findings of the work on aggregation and multiresolution analysis.

As for the relation with the revealing work in the frequency domain, casting the problem in the time domain achieves two objectives for us. First, it may assist applicability given the average user's familiarity with classical time series methods in the time domain. Second, the factor structures in the literature are in the time domain. Preserving time, while still capturing horizon effects, allows us to relate to existing specifications and facilitate economic interpretability (c.f., e.g., Section 8).

We conclude this section by emphasizing that the frequency-specific nature of our quantities of risk (the spectral betas) is suggestive of the potential for frequency-dependent prices of risk. In this sense, our approach may be viewed as, also, complementary to the work of Dew-Becker and Giglio (2016) who study frequency-based decompositions of the prices of risk.¹⁰

4. Spectral factor models: a formalization

For the purpose of extracting orthogonal components (with cycles of different length) of both the factor and the excess asset returns, we make use of the extended Wold representation of a covariance-stationary process in Bandi, Perron, Tamoni, and Tebaldi (2019) and Ortu, Severino, Tamoni, and Tebaldi (2020).

Without loss of generality, we focus on the bivariate case in order to present ideas without complicating notation unnecessarily. After reviewing the extended Wold representation, we apply it to construct spectral factor models and to derive a novel representation of the beta of a given bivariate process as a linear combination of the betas of its orthogonal (frequency-specific) components.

4.1. *The extended Wold representation*

Let $\mathbf{x} = \{(y_t, x_t)^\top\}_{t \in \mathbb{Z}}$ be a covariance-stationary bivariate process. For simplicity, assume the process is mean zero. Adding a mean would, of course, just amount to adding a constant term to its Wold representation below.

Define the white noise process $\boldsymbol{\varepsilon} = \{(\varepsilon_t^1, \varepsilon_t^2)^\top\}_{t \in \mathbb{Z}}$ such that $\mathbb{E}[\boldsymbol{\varepsilon}] = 0$ and $\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top] = \Sigma_\varepsilon$, where Σ_ε is a covariance matrix of dimension 2×2 . For any t in $\mathbb{Z}$, $\mathbf{x}$ satisfies the following

¹⁰Dew-Becker and Giglio (2016) evaluate conditional asset pricing models. Extending spectral factor models to a conditional setting is an interesting direction for future research.

Wold representation:

$$\begin{pmatrix} y_t \\ x_t \end{pmatrix} = \sum_{k=0}^{\infty} \begin{pmatrix} \alpha_k^1 & \alpha_k^2 \\ \alpha_k^3 & \alpha_k^4 \end{pmatrix} \begin{pmatrix} \varepsilon_{t-k}^1 \\ \varepsilon_{t-k}^2 \end{pmatrix} = \sum_{k=0}^{\infty} \boldsymbol{\alpha}_k \boldsymbol{\varepsilon}_{t-k}, \quad (12)$$

with $\sum_{k=0}^{\infty} \text{tr}^{1/2}(\boldsymbol{\alpha}_k^T \boldsymbol{\alpha}_k) < \infty$ and $\boldsymbol{\alpha}_0 = I_2$, a 2×2 identity matrix, where the equality is in the $\mathbb{L}^2$ -norm.

Define $\mathbf{x}^{(j)}$ as the j -th component of the process $\mathbf{x}$. Straightforward aggregation of the shocks to the system leads to the equivalent *extended* Wold representation

$$\begin{pmatrix} y_t \\ x_t \end{pmatrix} = \sum_{j=1}^{\infty} \sum_{k=0}^{\infty} \boldsymbol{\Psi}_k^{(j)} \boldsymbol{\varepsilon}_{t-k2^j}^{(j)} = \sum_{j=1}^{\infty} \mathbf{x}_t^{(j)} \quad (13)$$

in which, for any $j \in \mathbb{N}$, the 2×2 matrices $\boldsymbol{\Psi}_k^{(j)}$ are the unique discrete Haar transforms (DHT) of the original Wold coefficients, i.e.,

$$\boldsymbol{\Psi}_k^{(j)} = \frac{1}{\sqrt{2^j}} \begin{pmatrix} \sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+i} & \sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+2^{j-1}+i} \end{pmatrix}, \quad (14)$$

and the 2×1 vectors $\boldsymbol{\varepsilon}_t^{(j)}$ are the DHTs of the original Wold shocks, i.e.,

$$\boldsymbol{\varepsilon}_t^{(j)} = \frac{1}{\sqrt{2^j}} \begin{pmatrix} \sum_{i=0}^{2^{j-1}-1} \varepsilon_{t-i} & \sum_{i=0}^{2^{j-1}-1} \varepsilon_{t-2^{j-1}-i} \end{pmatrix}. \quad (15)$$

We emphasize that the shocks to the j -th component coincide with the j -th component of the shocks to the raw series. This is an important property which speaks to a coherent link between the innovations to the raw series and the innovations to its components.

The representation in Eq. (13) amounts to an orthogonal decomposition of the original bivariate series into uncorrelated components (for every t) operating over alternative frequencies (denoted by the index j). This is easily seen. Take the first and the second frequency, i.e., $j = 1$ and $j = 2$, and notice that the corresponding shocks are defined as follows:

$$\boldsymbol{\varepsilon}_t^{(1)} = \begin{pmatrix} \frac{\varepsilon_t^1 - \varepsilon_{t-1}^1}{\sqrt{2}} \\ \frac{\varepsilon_t^2 - \varepsilon_{t-1}^2}{\sqrt{2}} \end{pmatrix}, \quad \boldsymbol{\varepsilon}_{t-2}^{(1)} = \begin{pmatrix} \frac{\varepsilon_{t-2}^1 - \varepsilon_{t-3}^1}{\sqrt{2}} \\ \frac{\varepsilon_{t-2}^2 - \varepsilon_{t-3}^2}{\sqrt{2}} \end{pmatrix}, \quad \dots$$

and

$$\boldsymbol{\varepsilon}_t^{(2)} = \begin{pmatrix} \frac{(\varepsilon_t^1 + \varepsilon_{t-1}^1) - (\varepsilon_{t-2}^1 + \varepsilon_{t-3}^1)}{\sqrt{4}} \\ \frac{(\varepsilon_t^2 + \varepsilon_{t-1}^2) - (\varepsilon_{t-2}^2 + \varepsilon_{t-3}^2)}{\sqrt{4}} \end{pmatrix}, \quad \boldsymbol{\varepsilon}_{t-4}^{(2)} = \begin{pmatrix} \frac{(\varepsilon_{t-4}^1 + \varepsilon_{t-5}^1) - (\varepsilon_{t-6}^1 + \varepsilon_{t-7}^1)}{\sqrt{4}} \\ \frac{(\varepsilon_{t-4}^2 + \varepsilon_{t-5}^2) - (\varepsilon_{t-6}^2 + \varepsilon_{t-7}^2)}{\sqrt{4}} \end{pmatrix}, \quad \dots$$

The coefficient matrices have the same structure, e.g., for $j = 1$ we have

$$\Psi_0^{(1)} = \begin{pmatrix} \frac{\alpha_0^1 - \alpha_1^1}{\sqrt{2}} & \frac{\alpha_0^2 - \alpha_1^2}{\sqrt{2}} \\ \frac{\alpha_0^3 - \alpha_1^3}{\sqrt{2}} & \frac{\alpha_0^4 - \alpha_1^4}{\sqrt{2}} \end{pmatrix}, \quad \Psi_1^{(1)} = \begin{pmatrix} \frac{\alpha_2^1 - \alpha_3^1}{\sqrt{2}} & \frac{\alpha_2^2 - \alpha_3^2}{\sqrt{2}} \\ \frac{\alpha_2^3 - \alpha_3^3}{\sqrt{2}} & \frac{\alpha_2^4 - \alpha_3^4}{\sqrt{2}} \end{pmatrix}, \quad \dots$$

and, for $j = 2$,

$$\Psi_0^{(2)} = \begin{pmatrix} \frac{(\alpha_0^1 + \alpha_1^1) - (\alpha_2^1 + \alpha_3^1)}{\sqrt{4}} & \frac{(\alpha_0^2 + \alpha_1^2) - (\alpha_2^2 + \alpha_3^2)}{\sqrt{4}} \\ \frac{(\alpha_0^3 + \alpha_1^3) - (\alpha_2^3 + \alpha_3^3)}{\sqrt{4}} & \frac{(\alpha_0^4 + \alpha_1^4) - (\alpha_2^4 + \alpha_3^4)}{\sqrt{4}} \end{pmatrix}, \quad \Psi_1^{(2)} = \begin{pmatrix} \frac{(\alpha_4^1 + \alpha_5^1) - (\alpha_6^1 + \alpha_7^1)}{\sqrt{4}} & \frac{(\alpha_4^2 + \alpha_5^2) - (\alpha_6^2 + \alpha_7^2)}{\sqrt{4}} \\ \frac{(\alpha_4^3 + \alpha_5^3) - (\alpha_6^3 + \alpha_7^3)}{\sqrt{4}} & \frac{(\alpha_4^4 + \alpha_5^4) - (\alpha_6^4 + \alpha_7^4)}{\sqrt{4}} \end{pmatrix}, \quad \dots$$

It is now straightforward to verify that the shocks associated with the first and the second component are white noise processes (with variances Σ_ε) when defined on the supports $S_t^{(1)} = \{t - k2 : k \in \mathbb{Z}\}$ and $S_t^{(2)} = \{t - k2^2 : k \in \mathbb{Z}\}$. The components $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$ are orthogonal at all leads and lags. More generally, for generic components j and k , we have

$$\mathbb{E} \left[\mathbf{x}_{t-m2^j}^{(j)} \mathbf{x}_{t-n2^k}^{(k) \top} \right] = \mathbf{0} \quad \forall j \neq k, \quad \forall m, n \in \mathbb{N}_0, \quad \forall t \in \mathbb{Z}. \quad (16)$$

In essence, the bivariate time series $\mathbf{x}$ has been re-expressed as an infinite sum of Wold components, i.e., $\mathbf{x}_t = \sum_{j=1}^{\infty} \mathbf{x}_t^{(j)}$. Each Wold component has a Wold representation with white noise shocks suitably defined on an enlarging support $S_t^{(j)} = \{t - k2^j : k \in \mathbb{Z}\}$. Therefore, the generic j -th Wold component will capture cycles with length between 2^{j-1} and 2^j periods. The index j is, therefore, associated with the j -th frequency/scale.¹¹

4.2. A novel factor structure and a β representation

Linear factor models are routinely used in financial applications. The capital asset pricing model (CAPM) introduced by Sharpe (1964) and Lintner (1965), the arbitrage pricing theory (Ross, 1976; Connor and Korajczyk, 1986) and the Fama-French (1993) three-factor model are, to name just a few, classical textbook specifications. Importantly, as implied by a direct comparison of Eq. (1) with Eq. (4), the beta coefficients in all of these models are constant across frequencies.

The orthogonal representation in Eq. (13) allows us to decompose the factor into components (i.e., $x_t^{(j)}$ with $j \geq 1$) with various degrees of periodicity. The representation leads

¹¹When using *monthly* data, as we do unless otherwise stated, we refer to Table 1 for a translation of the scales into time horizons.

to the following *spectral factor model*:

$$y_t = \alpha + \sum_{j=1}^{\infty} \beta^{(j)} x_t^{(j)} + u_t. \quad (17)$$

In the model, the betas (indexed by j) associated with individual Wold components measure frequency-specific dependence and define our notion of *spectral betas*.

We emphasize that, for the purpose of modeling as well as in implementations, one does not need to truncate the infinite sum up to a large, but finite, J . This is easy to see. In agreement with our example in Section 2, the series x_t can be broken down into any number of components capturing fluctuations over specific ranges of frequencies. For example, one may work with $x_t^{<2^{j^*}}$ and $x_t^{>2^{j^*}}$, where $x_t^{>2^{j^*}} = x_t - x_t^{<2^{j^*}}$, for some $j^* \geq 1$. As an additional example, one may also work with a three-component representation with $x_t^{<2^{j^*}}$, $x_t^{(2^{j^*}, 2^{j^\diamond})}$ and $x_t^{>2^{j^\diamond}} = x_t - x_t^{<2^{j^*}} - x_t^{(2^{j^*}, 2^{j^\diamond})}$, where $j^\diamond > j^* \geq 1$. In both examples, the last (i.e., residual) component is always the original series x_t purged of the assumed higher-frequency components. Hence, the decomposition is always *exact* and can always be conducted for any $J \geq 1$.

In sum, we can always write

$$y_t = \alpha + \sum_{j=1}^J \beta^{(j)} x_t^{(j)} + \beta^{(J+1)} \pi_t^{(J)} + u_t, \quad (18)$$

with residual component $\pi_t^{(J)} = x_t - \sum_{j=1}^J x_t^{(j)}$.

Not only is the lack of truncation theoretically interesting, it is also empirically useful. Should one deem that the length of the data set is such that certain (lower) frequencies can not be studied reliably, the index J may be set to a lower number, thereby simply employing a residual term $\pi_t^{(J)}$ which represents a larger range of frequencies.

Interestingly, the beta associated with the residual component ($\beta^{(J+1)}$) can be readily interpreted. Next, we discuss how to do so. We begin by showing that the overall beta is a linear combination of betas on the Wold components.

Theorem 1 (A β representation.) Assume $\mathbf{x} = \{(y_t, x_t)^\top\}_{t \in \mathbb{Z}}$ satisfies Eq. (12). Define the spectral beta associated with frequency j as $\beta^{(j)} = \frac{\mathbb{E}[y_t^{(j)}, x_t^{(j)}]}{\mathbb{V}[x_t^{(j)}]}$. The overall beta would, therefore, conform with

$$\beta = \frac{\mathbb{C}[y_t, x_t]}{\mathbb{V}[x_t]} = \sum_{j=1}^{\infty} v^{(j)} \beta^{(j)}, \quad (19)$$

where $v^{(j)} = \frac{\mathbb{V}[x_t^{(j)}]}{\mathbb{V}[x_t]}$.

Proof. Immediate, given the representation in Eq. (13) and its properties.

In essence, the traditional beta can always be expressed as a weighted average of spectral betas with weights directly related to the *relative* informational content of the corresponding frequencies. The latter is defined by $v^{(j)} = \frac{\mathbb{V}[x_t^{(j)}]}{\mathbb{V}[x_t]}$.

Similarly, the beta on a range of frequencies is a linear combination of the betas corresponding to the same frequencies, a finding which provides direct interpretation of, e.g., the beta on the residual component $\pi_t^{(J)}$ for any choice of J . The following corollary provides the corresponding result.

Corollary 1. Assume $\mathbf{x} = \{(y_t, x_t)^\top\}_{t \in \mathbb{Z}}$ satisfies Eq. (12). Define the spectral beta associated with frequency j as $\beta^{(j)} = \frac{\mathbb{E}[y_t^{(j)}, x_t^{(j)}]}{\mathbb{V}[x_t^{(j)}]}$. The beta over $j \in (j^*, j^\circ)$ would, therefore, conform with

$$\beta^{(j^*, j^\circ)} = \sum_{j \in (j^*, j^\circ)} v^{(j)} \beta^{(j)}, \quad (20)$$

where $v^{(j)} = \frac{\mathbb{V}[x_t^{(j)}]}{\mathbb{V}[\sum_{j \in (j^*, j^\circ)} x_t^{(j)}]}$.

Proof. Immediate, given Theorem 1.

Next, we discuss identification of the components, before turning to the empirical analysis.

5. Identification

In order to operationalize the extended Wold representation in Eq. (13), we first need to compute the Wold coefficients α_k .

To this end, we write the vector time series of interest as $\mathbf{x}_t = (y_t, \tilde{\mathbf{x}}_t)^\top \in \mathbb{R}^k$, where $y_t \in \mathbb{R}$ and $\tilde{\mathbf{x}}_t \in \mathbb{R}^{k-1}$. For clarity, one may view the scalar process y_t as the return on a generic asset and the process $\tilde{\mathbf{x}}_t$ as either a process inclusive of multiple factors or a process inclusive of factors and additional state variables. We will be explicit about the specification we adopt in empirical work in the next subsection.

We assume $\mathbf{x}_t$ follows a linear vector autoregressive (VAR) process of order p (VAR(p)) of the form:

$$\mathbf{x}_t = A_1 \mathbf{x}_{t-1} + \dots + A_p \mathbf{x}_{t-p} + \boldsymbol{\varepsilon}_t, \quad (21)$$

where the A_i s, with $i = 1, \dots, p$, are $k \times k$ parameter matrices and the error process, $\boldsymbol{\varepsilon}_t$, is a k -dimensional zero-mean white noise process with covariance matrix $\mathbb{E}(\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t^\top) = \Sigma_\varepsilon$.

As is well-known, the VAR(p) process in Eq. (21) can be written as a kp -dimensional VAR(1) process by stacking p consecutive $\mathbf{x}_t$ variables in a $(kp \times 1)$ -dimensional vector $\mathbf{X}_t = (\mathbf{x}_t^\top, \mathbf{x}_{t-1}^\top, \dots, \mathbf{x}_{t-p+1}^\top)^\top$. Thus, we have

$$\mathbf{X}_t = A\mathbf{X}_{t-1} + U_t, \quad (22)$$

where the matrix A is sometimes referred to as the “companion matrix” of the VAR(p) process. It is well-known that $\mathbf{X}_t$ is stable if $\det(I_{kp} - Az) \neq 0 \forall |z| \leq 1$. It is also easy to see that this condition is equivalent to assuming that all of the eigenvalues of A have modulus less than one.

Under standard assumptions, a stable VAR has a time-invariant mean and a time-invariant variance/covariance matrix and is, therefore, covariance stationary (see, e.g., Lutkepohl, 2005, Chapter 2). Put differently, the process admits a Wold representation. Indeed, the stability of the process ensures the existence of the inverse VAR operator $(I_{kp} - AL)^{-1} = \sum_{k=0}^{\infty} \alpha_k L^k$, where $\alpha_k = A^k$. As a result, one can obtain the following Wold moving average representation of $\mathbf{X}_t$:

$$\mathbf{X}_t = \sum_{k=0}^{\infty} \alpha_k U_{t-k}. \quad (23)$$

In light of this discussion, identification of the Wold components proceeds as follows. We assume a VAR(p) process for $\mathbf{x}_t$, obtain its companion representation in Eq. (22) and estimate the resulting VAR(1) model using unrestricted least squares. We then derive the Wold coefficients by virtue of the restriction $\alpha_k = A^k$ and use them in Eq. (14) to evaluate the coefficients of the extended Wold representation. Finally, employing the residuals from the VAR(1) along with Eq. (15), we obtain the frequency-specific innovations of the extended Wold representation.

5.1. *Incorporating economic restrictions: identification in practice*

In all of our subsequent implementations,¹² we account for predictability in market returns. We do so by setting the vector $\tilde{\mathbf{x}}_t$ as being the market excess return followed by three state variables: the yield spread between long-term and short-term bonds (measured as the difference between the ten-year constant maturity bond yield and the yield on short-term notes, in annualized percentage points), the market’s price-dividend ratio (measured as the logarithmic ratio of the CRSP price index and a one-year moving average of dividends) and the small-stock value spread (measured as the difference between the logarithmic book-to-market ratios of small value and small growth stocks). Our use of these state variables in

¹²Appendix A.3 is the only exception.

the VAR is guided, e.g., by the work of Campbell and Vuolteenaho (2004).

More specifically, for each asset return y we estimate a VAR(p) with block exogeneity defined as follows:

$$\begin{aligned} y_t &= a_y + A_{1,y}Y_{t-1} + A_{2,y}\tilde{\mathbf{X}}_{t-1} + \varepsilon_t^1 \\ \tilde{\mathbf{x}}_t &= a_x + A_{1,x}Y_{t-1} + A_{2,x}\tilde{\mathbf{X}}_{t-1} + \varepsilon_t^2, \end{aligned}$$

where $Y_{t-1} = [y_{t-1}, y_{t-2}, \dots, y_{t-p}]^\top$ and $\tilde{\mathbf{X}}_{t-1} = [\tilde{\mathbf{x}}_{t-1}^\top, \tilde{\mathbf{x}}_{t-2}^\top, \dots, \tilde{\mathbf{x}}_{t-p}^\top]^\top$.¹³

Importantly, we set $A_{1,x} = 0$. In other words, y does not Granger cause $\tilde{\mathbf{x}}$.¹⁴ Economically, we are letting the extraction of the market components (the excess market return is the first element in $\tilde{\mathbf{x}}$) be independent of the test asset(s) being considered (we recall that y is a specific asset's return).

6. An empirical evaluation of the β representation

The goal of this section is to provide a preliminary empirical illustration of the proposed beta representation. We also briefly report on the representation's potential for understanding cross-sectional variation in expected returns, the object of our analysis in Section 7. We do so for monthly portfolios returns.¹⁵

We run market model-style regressions on two assets: a high book-to-market “value” portfolio and a low book-to-market “growth” portfolio.¹⁶ All returns are in excess of the risk-free rate. The results are

$$\begin{aligned} R_{value,t} &= \alpha + 1.211 \times R_{m,t} + u_t, \quad \mathbb{R}^2 = 0.78, \\ &\quad (t\text{-stat} = 20.39) \end{aligned}$$

$$\begin{aligned} R_{growth,t} &= \alpha + 0.974 \times R_{m,t} + u_t, \quad \mathbb{R}^2 = 0.73. \\ &\quad (t\text{-stat} = 27.50) \end{aligned}$$

The regressions show the well-known result that these portfolios have a market beta around one. Since the volatility of the excess market return series in our sample is 15.25% per annum, the beta estimates imply a covariance equal to $1.211 \times (15.25/\sqrt{12})^2 = 23.276$ for

¹³The lag length p of the VAR is set to 18. Results are robust to alternative choices around 18.

¹⁴Hansen, Heaton, and Li (2008) adopt a similar restriction. In particular, they restrict the dividend process of a portfolio not to Granger-cause aggregate consumption.

¹⁵We examine daily portfolio returns in Appendix A.3.

¹⁶We use the first and the tenth value-weighted decile portfolios formed on book-to-market from Kenneth French's data library. The data are from January 1967 through December 2018.

Interpretation of the scales

Scale	Frequency resolution
$j = 1$	1 – 2 months
$j = 2$	2 – 4 months
$j = 3$	4 – 8 months
$j = 4$	8 – 16 months
$j = 5$	16 – 32 months
$j = 6$	32 – 64 months
$j > 6$	> 64 months

Table 1: The table provides the interpretation of each scale in terms of monthly time spans.

value and $0.974 \times (15.25/\sqrt{12})^2 = 18.876$ for growth.

The top two rows of Table 2 focus on covariance decomposition. We obtain the Wold coefficients, the Wold residuals and, ultimately, the Wold components, as described in Subsection 5.1. Using the Wold components, we then compute $\hat{\mathbb{C}}(\hat{R}_m^{(j)}, \hat{R}_p^{(j)})$, where $p = \{value, growth\}$ with $j = 1, \dots, 6$, and $\hat{\mathbb{C}}(\hat{\pi}_m^{(6)}, \hat{\pi}_p^{(6)})$. The interpretation of the scales in terms of monthly horizons is contained in Table 1.

The last column of Table 2 shows that, by summing the frequency-specific covariances, we are able to recover the overall covariances reported above. The small differences between the covariances on the full sample and the sum of the frequency-specific covariances are due to the different sample lengths induced by the initializations of each spectral component (which require a minimum of 2^j observations) and estimation uncertainty.

The bottom rows in Table 2 illustrate the beta decomposition. We observe a widening of the dispersion in the betas associated with the value and growth portfolios as we move from high frequencies (low j values) to low frequencies (high j values).

The last column, again, shows accurate reconstruction of the overall betas. As made explicit in Theorem 1, in order to obtain the overall beta, it is important to re-weight the frequency-specific betas:

$$\sum_{j=1}^7 \hat{v}^{(j)} \hat{\beta}^{(j)} \quad \text{with} \quad \hat{v}^{(j)} = \frac{\hat{\mathbb{V}}(\hat{R}_m^{(j)})}{\hat{\mathbb{V}}(\hat{R}_m)}.$$

Next, we explore the *orthogonality* property of the proposed representation. To simplify the notation, we bundle together the frequencies below (about) one year and above (about) one year. Specifically, for each return series, we sum all of the components up to scale

	$j = 1$	$j = 2$	$j = 3$	$j = 4$	$j = 5$	$j = 6$	$j > 6$	$\sum_{j=1}^7 \hat{C}$
Spectral covariances								
Value	8.362	8.935	4.055	1.548	0.866	0.215	0.117	24.097
Growth	7.392	6.549	2.802	1.174	0.634	0.147	0.052	18.749
Spectral betas and variance weights								
	$j = 1$	$j = 2$	$j = 3$	$j = 4$	$j = 5$	$j = 6$	$j > 6$	$\sum_{j=1}^7 \hat{v}^{(j)} \hat{\beta}^{(j)}$
Value	1.167	1.215	1.341	1.151	1.229	0.962	0.901	1.208
weight (rel. variance)	0.419	0.448	0.203	0.078	0.043	0.011	0.006	
Growth	1.032	0.891	0.927	0.873	0.900	0.656	0.402	0.940
weight (rel. variance)	0.371	0.328	0.141	0.059	0.032	0.007	0.003	

Table 2: Book-to-market sorted portfolios: covariance and beta decomposition. The panels provide spectral covariances, spectral betas and relative variance weights associated with seven frequencies for a value and a growth portfolio. We decompose the portfolios (excess) returns and the market (excess) returns into six orthogonal components in addition to a residual term $\pi_t^{(6)}$, as discussed in Section 5. All returns are in percent (i.e., the raw covariances are multiplied by 10^4).

4 (included) and call this new component the “high-frequency” (HF) component.¹⁷ Similarly, for each return series, we sum all of the components higher than scale 4 and call this new component the “low-frequency” (LF) component. We then run the following simple regressions:

$$\begin{aligned}
 R_{p,t}^{HF} &= \alpha + \beta_p^{HF} \times R_{m,t}^{HF} + u_t, \\
 R_{p,t}^{LF} &= \alpha + \beta_p^{LF} \times R_{m,t}^{LF} + u_t.
 \end{aligned}$$

By the orthogonality of the components, the corresponding multiple regression, i.e.,

$$R_{p,t} = \alpha + \beta_p^{HF} \times R_{m,t}^{HF} + \beta_p^{LF} \times R_{m,t}^{LF} + u_t,$$

should deliver analogous beta estimates. We emphasize that these specifications are the empirical analogues of the regressions in Eq. (9), Eq. (10) and Eq. (4) of Section 2. Table 3 displays the results.

Consistent with theory regarding the orthogonality of the components, we find that multiple and simple regressions deliver similar estimates. Once more, we observe a wider

¹⁷The cut-off $j = 4$ with monthly data corresponds to $2^4/12 = 1.\bar{3}$ years.

	Simple regression		Multiple regression	
	β^{LF}	β^{HF}	β^{LF}	β^{HF}
Value	1.264 (9.088)	1.219 (18.545)	1.289 (7.382)	1.232 (18.423)
Growth	0.839 (7.563)	0.957 (25.181)	0.851 (6.710)	0.962 (23.187)

Table 3: Book-to-market sorted portfolios. We run simple and multiple regressions on high-frequency and low-frequency market components for a value and a growth portfolio. We decompose the portfolios (excess) returns and the market (excess) returns into six orthogonal components in addition to a residual term $\pi_t^{(6)}$, as discussed in Section 5. We obtain the high-frequency market component by summing the market components up to scale $j = 4$ (included). The sum of the remaining market components defines the low-frequency market component. The table provides estimated betas and t -statistics.

dispersion in the betas at low frequency, which is consistent with previous results in Table 2. Our conclusions would not change if one were to choose a different cut-off j , except that the spread between high-frequency betas and low-frequency betas would be more or less exacerbated. The pricing implications of such spread are the subject of the next section.

7. Cross-sectional pricing: the role of spectral betas

While any factor may, in principle, be separated into components, we will continue with the analysis in the previous section and place emphasis on the market. There are key reasons for this choice. First, because the market is a textbook factor, economic interpretability will be preserved. Second, in light of the notorious shortcomings of the market factor in empirical cross-sectional asset pricing, our study of the relative pricing content of individual market components will have a natural benchmark, the aggregate market factor itself.¹⁸ Finally, the market is a factor in the vast majority of cross-sectional pricing models. Hence, the potential of the methods for dimensionality reduction may only be meaningfully studied by focusing, as we do, on components of a broadly-utilized factor. In sum, our focus on the market gives us discipline. Consistent with these observations, Harvey, Liu, and Zhu (2016) argue that the unstructured search for the “best” factor model may run into an overwhelming multiple comparison problem and fully-specified models, like the CAPM, should be used to guide meaningful searches.

To set the stage for our formal analysis, it is natural to begin with a descriptive take

¹⁸The work of Harvey and Liu (2019) represents a dissenting voice in the asset pricing literature, one which re-attributes - as we do by exploring frequency - a first-order role to the market factor.

on the most classical cross section in applied asset pricing work, namely the 25 size and book-to-market sorted portfolios of Fama and French. Panel (a) in Figure 1 reports average realized returns for each size/value quintile. The conclusion is standard: average returns decrease with size (the only exception being the first value quintile) and increase with value. For the market factor to price these portfolios, the corresponding CAPM betas would have to align with the reported average returns. Such an alignment, however, has been elusive, thereby generally leading to lack of support for the CAPM in the data.

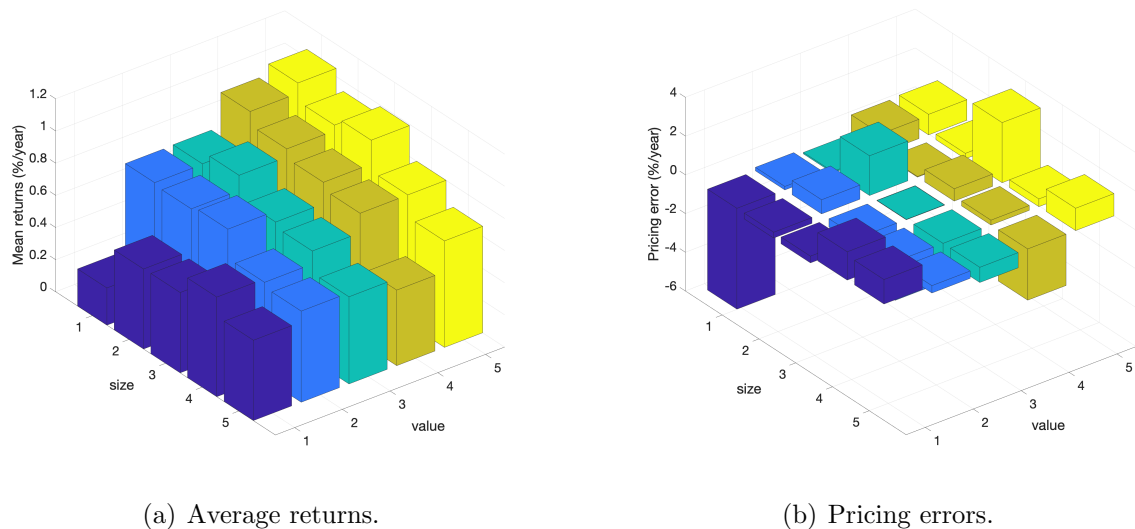


Fig. 1. **25 Fama-French book-to-market and size sorted portfolios.** We report average realized returns on the 25 Fama-French portfolios sorted on size and book-to-market (Panel (a)) along with pricing errors from a spectral CAPM model (Panel (b)). The data are monthly from January 1967 through December 2018.

Rather than focusing on the overall market betas, we decompose (as discussed in Section 5) the portfolios (excess) returns and the market (excess) returns into six orthogonal components (plus a residual term $\pi_t^{(6)}$ defined in Subsection 4.2) and calculate spectral covariances/betas corresponding to the components.¹⁹ The spectral covariances are reported in Figure 2. We observe that the spectral covariances have a marked size tilt, in particular over economic cycles between 2 and 4 months ($j = 2$) and 4 and 8 months ($j = 3$). As we move towards lower frequencies, we witness a rotation towards value which leads to the $j = 6$ spectral covariances being nicely aligned with average realized returns both in the size and

¹⁹The six market components and the residual term have decreasing empirical variances equal to 7.701, 6.921, 3.845, 2.308, 0.677, 0.289 and 0.147. The decreasing pattern of the variances is typical of processes with very low autocorrelation. In fact, the first-order monthly autocorrelation of the market return is 0.1 in our data. As expected, while the variances of the components and the residual term are decreasing, their monthly autocorrelations are increasing and equal to -0.471 , 0.228 , 0.596 , 0.812 , 0.909 , 0.947 and 0.965 .

in the value dimension. These spectral covariances correspond to cycles between 32 months and 64 months. In other words, they correspond to business-cycle fluctuations.

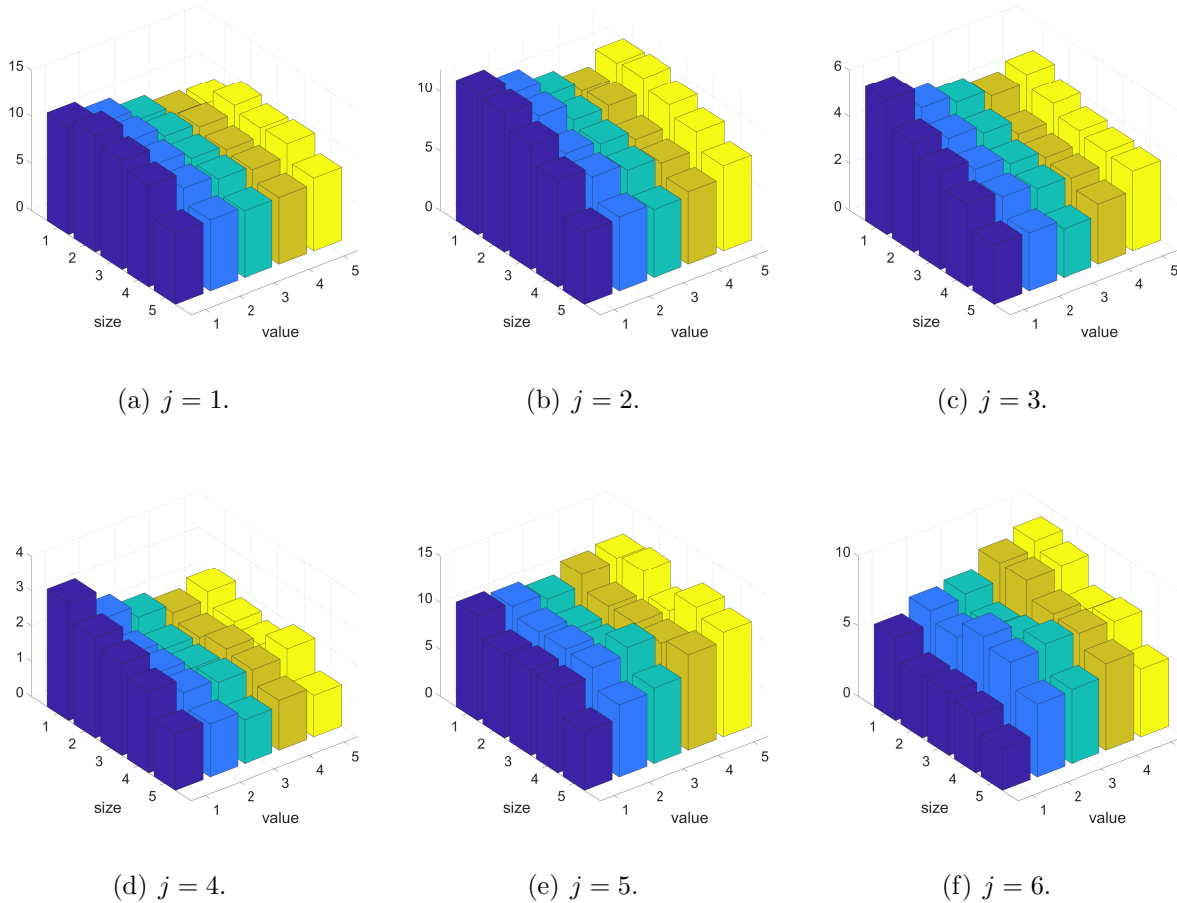
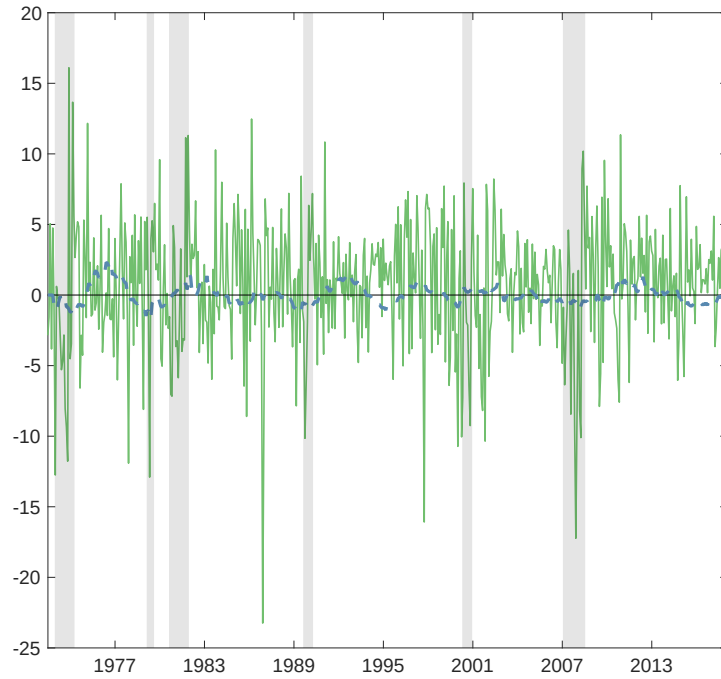
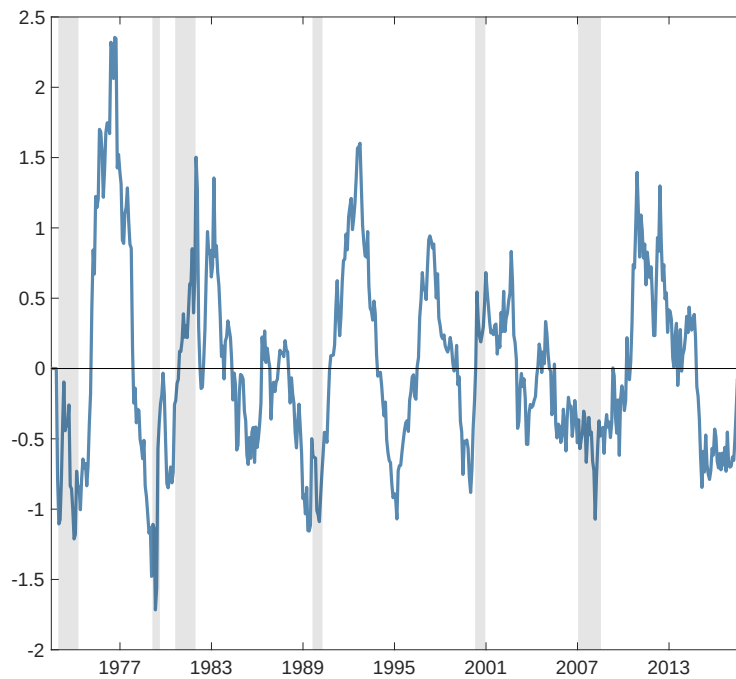


Fig. 2. **25 Fama-French book-to-market and size sorted portfolios.** We report spectral market covariances ($\widehat{C}^{(j)}$) associated with alternative scales $j = 1, \dots, J$, where the generic scale j captures fluctuations between $2^{(j-1)}$ and 2^j months. We set $J = 6$. The data are monthly from January 1967 through December 2018.

In what follows, we will sometimes refer to the $j = 6$ market component, whose graphical representation is provided in Figure 3, as a “business-cycle” component. We will also sometimes define the corresponding $j = 6$ spectral covariances/betas as “business-cycle” covariances/betas. The business-cycle market component is slow moving and explains a small fraction of the overall market variance. As made clear by the beta representation in Theorem 1, because the variance ratio is low, the aggregate beta hides the pricing signal contained in the business-cycle spectral beta. Explicit separation of this “signal” from the “noise” at alternative frequencies is central to our approach.



(a) Market returns and business-cycle component at scale $j = 6$.



(b) Business-cycle component at scale $j = 6$.

Fig. 3. **The business-cycle component of the market return process, $\hat{R}_{m,t}^{(6)}$.** Panel (a) plots the market returns $R_{m,t}$ (solid line) against the estimated component at scale $j = 6$, $\hat{R}_{m,t}^{(6)}$ (dashed line). Panel (b) zooms in on the business-cycle market component in Panel (a). The data are monthly from January 1967 through December 2018.

We now turn to a formal analysis of the pricing ability of the business-cycle market component for the cross section of 25 size and book-to-market portfolios²⁰ as well as for an array of richer cross sections recently employed in the asset pricing literature. As emphasized, because we are interested in the potential of spectral factor models for both dimensionality reduction and economic interpretability, focusing on a single, business-cycle, component of the market return process is natural.

Table 4 reports the outcome of second-stage cross-sectional regressions of excess portfolio returns on business-cycle covariances. The table provides prices of risk, root mean squared errors (RMSE), mean absolute pricing errors (MAPE) and $\mathbb{R}^2$ s for six sets of test assets: the 25 Fama-French size and book-to-market portfolios, the 25 Fama-French size and operating profitability portfolios, the 25 Fama-French size and investment portfolios, the universe of French anomalies to which we add “quality-minus-junk” (Asness, Frazzini, and Pedersen, 2019) and “betting-against-beta” (Frazzini and Pedersen, 2014) in agreement with the recent work of Ehsani and Linnainmaa (2019),²¹ 24 anomalies from Hou et al. (2020)²² and the ten duration portfolios of Weber (2018). We focus on characteristic-managed portfolios, since factor structures are likely to be more stable for portfolios than for stocks, i.e., the factor loadings are less likely to change over time, in addition to frequency (see, e.g., the discussion in Kelly, Pruitt, and Su, 2019, and Lettau and Pelger, 2020).

The price of risk associated with the business-cycle market component is stable across test assets and hovers around a mean value of 3. We report both classical Fama and MacBeth (1973) standard errors (in parenthesis) and standard errors robust to model misspecification as in the recent work of Kan et al. (2013). Even the latter, which are understandably larger than the former for all cross sections, imply statistical significance of the price of risk at conventional levels. In light of the one-factor nature of the assumed spectral factor model (in which the only source of systematic risk is a slow-moving component of the market), the $\mathbb{R}^2$ s are particularly large. Whether they are large, in statistical terms, as compared to existing multi-factor models is the object of our subsequent analysis.

In unreported results, we have also restricted the zero-beta rate to be the same as the risk-free rate by constraining the intercept to be equal to zero. In this case, the $\mathbb{R}^2$ - which

²⁰A preliminary look at pricing errors is contained in Figure 1, Panel (b).

²¹The anomalies are those in Table 1 of Ehsani and Linnainmaa (2019) but we exclude momentum, to which we devote a dedicated discussion in what follows, and liquidity, due to the shorter associated sample size. Including liquidity would, however, not modify our findings. These anomalies lead to 24 portfolios. For all of them, with the exception of “quality-minus-junk” and “betting-against-beta”, we consider two portfolios, the top and the bottom decile portfolios, resulting in 22 portfolios. The remaining two portfolios are “quality-minus-junk” and “betting-against-beta” portfolios constructed, as is customary in the literature, as long/short portfolios.

²²Because we work with the top and the bottom decile portfolios for each anomaly, the 24 anomalies result in 48 portfolios, as reported in Panel E of Table 4.

will be later used in pairwise tests of model misspecification - loses some of its appeal in that it is defined (to be contained between 0 and 1) with respect to the sum of squared values of the dependent variable (the mean excess returns, in our case) rather than with respect to the sum of squared deviations around the cross-sectional mean of the dependent variable. We find that the estimated values of the prices of risk do not change in any significant way. They simply increase (resp. decrease) mildly when the (generally insignificant) estimated intercepts in the unrestricted specifications of the models in Table 4 are positive (resp. negative). For portfolios sorted on size and book-to-market, size and profitability, size and investment and duration, the RMSE (MAPE) increase only slightly by 1.6% (0.09%), 6.6% (14%), 6% (−2%) and 4.5% (10%). The largest increases in RMSE and MAPE are for the 24 portfolios (38% and 26%, respectively) and for the 48 portfolios (27% and 18%, respectively).

The cross section of duration portfolios in Panel F of Table 4 is interesting for two reasons. First, Weber (2018) shows that some otherwise successful risk factors find it hard to explain this novel cross section. In contrast, we show that a slow-moving component of the market return process performs well. Second, several explanations have been put forward to reconcile the negative relation between duration and future returns. Lettau and Wachter (2007), for example, emphasize the importance of cash-flow risk, whereas Weber (2018) argues in favor of extrapolation bias affecting investors' forecasts of the long-term growth prospects of high-duration firms. Our explanation based on business-cycle risk is new, complements the previous literature, and agrees with the findings in Kojen, Lustig, and Van Nieuwerburgh (2017). These authors provide evidence that the returns to value and growth assets co-move with bond market factors forecasting future economic activity at business-cycle horizons. Both papers, our and Kojen et al. (2017), assign a central role to the business cycle as a priced state variable. They focus on book-to-market sorted stocks and maturity sorted bond portfolios, whereas we study business-cycle market risk on a broader set of equity characteristics. Their approach is event-based, our is frequency-based.

Table 5 contains pairwise $\mathbb{R}^2$ -based tests of model (mis)specification. In Panel A, we compare a single-factor model in which the sole factor is a business-cycle market component to a multi-factor model inclusive of seven spectral components, the classical CAPM, the Fama-French three-factor model (Fama and French, 1993), the Fama-French five-factor model (Fama and French, 2015) and the q4-factor model of Hou et al. (2015). The tests evaluate the statistical distance between the pairwise model-implied $\mathbb{R}^2$ s. Again, we allow for model misspecification as in Kan et al. (2013). All tests are non-nested with the exception of the first (column 1 in Panel A).²³ In Panel B, we instead compare the classical CAPM to the

²³The distinction between being “nested” and “non-nested” is important for purposes of implementation. We refer to the discussion in Kan et al. (2013) for details.

Table 4: Cross section of asset prices. The table reports the outcome of the cross-sectional regression $\overline{R}_i^e = \lambda_0 + \lambda^{(6)}\widehat{\mathbb{C}}_i^{(6)} + \xi_i$, where $\overline{R}_i^e$ is the mean excess return of portfolio i and $\widehat{\mathbb{C}}_i^{(6)}$ is the corresponding spectral market covariance at scale $j = 6$ estimated in the first-pass regression. We use monthly excess returns on the 25 Fama-French portfolios formed on size and book-to-market (Panel A), size and operating profitability (Panel B), size and investment (Panel C); in Panel D the model is estimated using 11 anomalies (two portfolios, top and bottom decile, for each anomaly) from Kenneth French’s website, to which we add “quality-minus-junk” and “betting-against-beta” factors; in Panel E we use 24 anomalies (two portfolios, top and bottom decile, for each anomaly) from Hou et al. (2020); in Panel F the model is estimated using monthly excess returns on the ten portfolios sorted on cash-flow duration from Weber (2018). Details on data are provided in Appendix A.1. The table reports the estimates of the factor risk price $\widehat{\lambda}^{(6)}$ and the constant term, Fama and MacBeth (1973) standard errors (in parentheses) and the model misspecification-robust standard errors of Kan et al. (2013) (in braces). The last column provides the $\mathbb{R}^2$ of the corresponding cross-sectional regression and its standard error. Estimated values which are significant at the 10% level, or lower, are in bold fonts. We also report the root mean square error (RMSE) and the mean absolute pricing error (MAPE) across all test assets. These are expressed as percentages per year. The data are monthly from January 1967 through December 2018.

Constant	$\lambda^{(6)}$	RMSE	MAPE	$\mathbb{R}^2$
Panel A: 25 Size and Book-to-Market Portfolios				
0.109 (0.245) {0.426}	3.039 (0.825) {1.478}	1.593	1.087	0.52 (0.23)
Panel B: 25 Size and Profitability Portfolios				
0.147 (0.228) {0.400}	2.840 (0.843) {1.313}	1.153	0.884	0.66 (0.21)
Panel C: 25 Size and Investment Portfolios				
0.226 (0.219) {0.326}	2.609 (0.946) {1.332}	1.710	1.185	0.38 (0.36)
Panel D: 24 Portfolios				
0.612 (0.185) {0.351}	2.434 (0.757) {1.241}	1.840	1.554	0.44 (0.28)
Panel E: 48 Portfolios from Hou et al. (2020)				
0.535 (0.249) {0.532}	3.557 (0.920) {2.048}	1.415	1.214	0.61 (0.21)
Panel F: 10 Duration Portfolios				
-0.152 (0.411) {1.069}	3.226 (1.066) {1.601}	1.588	1.234	0.83 (0.14)

Fama-French three-factor model, the Fama-French five-factor factor model and the q4-factor model of Hou et al. (2015). In this case, all tests are, of course, nested. We note that, because of the negative price of risk associated with the aggregate market factor, the $\mathbb{R}^2$ comparisons involving the duration portfolios in both panels are not economically meaningful. We will, therefore, not comment on the duration portfolios in the context of Table 5.

Panel B provides a natural starting point. While it is known that the CAPM is vastly outperformed by state-of-the-art multi-factor models in terms of $\mathbb{R}^2$ values, less is known about its relative performance when allowing for model misspecification (a notable exception is the work of Kan et al. (2013)). Panel B shows that misspecification-robust p -values associated with a test of equal $\mathbb{R}^2$ s lead to strong rejections of the null of equal $\mathbb{R}^2$ s in all cases. When compared to the CAPM, the Fama-French five-factor model and the q4-factor model behave similarly and considerably better than the Fama-French three-factor model, the divergence in performance between the three multivariate models being more limited in the pricing of size and book-to-market portfolios, as expected.

Having made these observations, we ask how a single-factor spectral CAPM model would fare as compared to a spectral factor model inclusive of all seven spectral components, the CAPM and the above-mentioned multi-factor models (see Panel A of Table 5). We emphasize that the difference between a spectral factor model with all seven components and the CAPM lies in the fact that the former does not impose restrictions on the risk quantities (the spectral betas being free across spectral components) whereas the classical CAPM forces risk to be the same across frequencies and sets it equal to the standard CAPM beta. We find that the single-factor spectral CAPM yields $\mathbb{R}^2$ values which are 30% (for the 25 size and investment portfolios) to 66% (for the 25 size and operating profitability portfolios) larger than the CAPM. In all cases (with the exception of the 25 size and operating profitability portfolios), misspecification robust p -values favor the single-factor spectral model at the 10% level. The economic and statistical separation between the performance of the multi-factors model and the single-factor spectral model is more limited. While all multi-factor models (including, by construction, the spectral factor model with all components) yield $\mathbb{R}^2$ values which are larger than that of the single-factor spectral model, the misspecification-robust p -values of the pairwise tests point to statistically insignificant differences at conventional levels. This is particularly true when employing the largest number of anomaly portfolios as test assets. With 48 anomaly portfolios, the difference in $\mathbb{R}^2$ values between the Fama-French five-factor model and the single-factor spectral factor model or the difference in $\mathbb{R}^2$ values between the q4-factor model and the single-factor spectral factor model is only 0.015 or 0.016. In both cases, the p -values are equal to 0.356, thereby resulting in a failure to detect any statistical

Table 5: Tests of equality of cross-sectional $\mathbb{R}^2$ s. The table presents pairwise tests of equality of the least-squares cross-sectional $\mathbb{R}^2$ s of several beta pricing models. The models include: (1) the spectral CAPM with a business-cycle component; (2) the CAPM with seven frequency components; (3) the traditional CAPM of Sharpe (1964) and Lintner (1965); (4) the Fama and French (FF3, 1993) three-factor model; (5) the Fama and French (FF5, 2015) five-factor model; (6) the Hou et al. (HXZ, 2015) four-factor model. The models are estimated using monthly returns on alternative cross sections (described in Table 4 and Appendix A.1) from January 1967 through December 2018. In Panel A (resp. Panel B) we report the differences between the $\mathbb{R}^2$ s of the benchmark spectral factor model based on the $j = 6$ business-cycle component (resp. the benchmark CAPM model) and the alternative models in the various columns along with the associated p -values (in parentheses) for the test $H_0 : \mathbb{R}^2_{benchmark} = \mathbb{R}^2_{alternative}$. The p -values are computed under the assumption that the models are potentially misspecified as in Kan et al. (2013).

Panel A: Spectral Factor Model Vs. Alternative Factor Models						
Benchmark Model		(1)	(2)	(3)	(4)	(5)
Spectral Factor Model	versus	7 freq.	CAPM	FF3	FF5	HXZ
25 Size-B/M Portfolios		-0.223 (0.348)	0.434 (0.072)	-0.132 (0.238)	-0.222 (0.172)	-0.183 (0.192)
25 Size-OP Portfolios		-0.201 (0.331)	0.660 (0.051)	0.152 (0.116)	-0.215 (0.169)	-0.226 (0.160)
25 Size-Inv Portfolios		-0.278 (0.287)	0.296 (0.225)	-0.139 (0.209)	-0.337 (0.105)	-0.336 (0.104)
24 Portfolios		-0.170 (0.36)	0.366 (0.098)	-0.175 (0.242)	-0.374 (0.122)	-0.332 (0.134)
48 Portfolios		-0.042 (0.374)	0.491 (0.069)	0.178 (0.167)	-0.015 (0.356)	-0.016 (0.356)
10 Duration Portfolios		-0.166 (0.403)	-0.07 (0.261)	-0.152 (0.146)	-0.163 (0.136)	-0.115 (0.191)
Panel B: CAPM Vs. Alternative Multi-Factor Models						
CAPM	versus			FF3	FF5	HXZ
25 Size-B/M Portfolios				-0.566 (0.002)	-0.656 (0.007)	-0.617 (0.002)
25 Size-OP Portfolios				-0.508 (0.012)	-0.875 (0.002)	-0.886 (0.002)
25 Size-Inv Portfolios				-0.435 (0.004)	-0.633 (0.006)	-0.632 (0.005)
24 Portfolios				-0.541 (0.003)	-0.740 (0.001)	-0.698 (0.002)
48 Portfolios				-0.313 (0.016)	-0.506 (0.003)	-0.507 (0.002)
10 Duration Portfolios				-0.082 (0.126)	-0.093 (0.263)	-0.045 (0.252)

difference between the three models when using this large set of test assets.²⁴

Different anomalies are, of course, not priced equally well. In Figure 4 we offer a visual representation of the pricing performance of the single-factor spectral model with the larger set of test assets, i.e., 48 portfolios representing both ends of 24 anomalies. The figure compares average realized returns to average returns implied by the model. Panel (a) includes all 24 anomalies and the resulting R^2 is a substantial value of 61%. We have already commented on the fact that the performance of either the Fama-French five-factor model or the q4-factor model of Hou et al. (2015) is virtually identical to what is reported in Panel (a). Panel (b) offers the same information but it excludes operating accruals, return on equity, the change in return on equity, composite equity issuance, total volatility and idiosyncratic volatility (the red portfolios in Panel (a)). Such an exclusion raises the R^2 to 73%. The reason is simple. Operating accruals, return on equity and changes in return on equity have inverted betas with respect to the business-cycle market component. Specifically the bottom operating-accrual portfolio should have a larger spectral beta than the top operating-accrual portfolio. The opposite is true. Similarly, the top return-on-equity portfolio or the top change-in-return-on-equity portfolio should have a larger beta than the corresponding bottom portfolios. Again, the opposite is true. Equity issuance, total volatility and idiosyncratic volatility have a large spread in terms of realized average returns between the top and the bottom decile but their spectral betas are relatively flat, albeit not inverted.

It is important to recognize that these anomalies are well-known to be extremely challenging even for state-of-the-art multi-factor models. Fama and French (2016) emphasize the inability of their five-factor model to price accruals and momentum (we will turn to a dedicated discussion about momentum in what follows). They also emphasize that large share-issuance and large volatility portfolios have low average realized returns which are hard to replicate given their model-implied risk. For the same portfolios, we witness some distance between the (large) model-implied expected returns and the average returns in the data. In agreement with Fama and French (2016), the distance is more limited for portfolios at the other end of the same anomalies, namely for the net-share-repurchase portfolios and for the low-volatility portfolios.

²⁴Because of our emphasis on the business cycle, we have also considered a comparison between the proposed spectral factor model and a model in which the factor is *ultimate consumption* (Parker and Julliard, 2005), a non-traded macro factor different from the traded return factors in all other specifications. *Ultimate consumption* is computed, in agreement with the original paper, as the growth rate in real per-capita nondurable consumption over 3 years. The test statistics (p -values) across portfolios are 0.113 (0.161) for size and book-to-market, 0.642 (0.035) for size and operating profitability, -0.062 (0.322) for size and investment, 0.021 (0.360) for the 24 portfolios, 0.298 (0.100) for the 48 portfolios, and -0.017 (0.360) for the duration portfolios. We conclude that a business-cycle market component performs either on par with ultimate consumption or better. The latter is the case for the size and profitability portfolios and for the 48 portfolios.

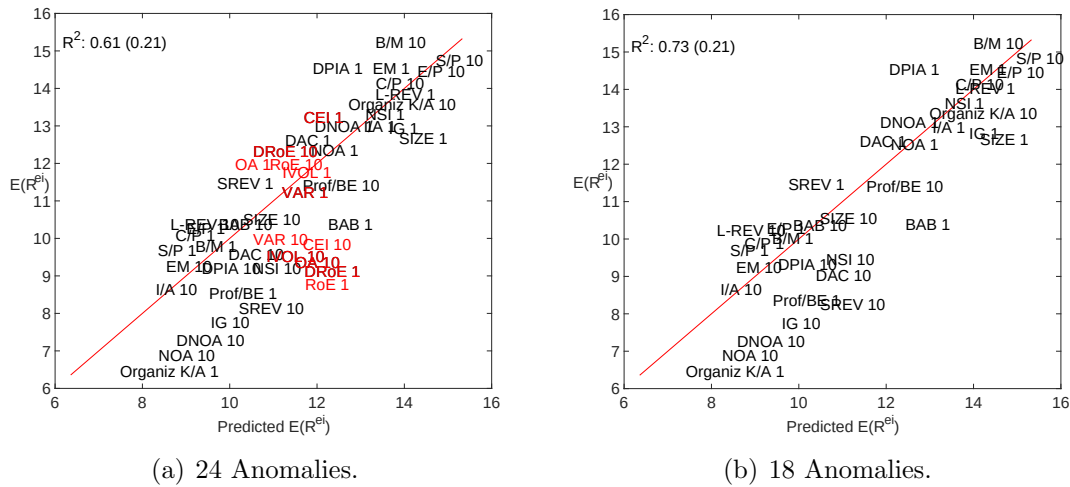


Fig. 4. **Cross-sectional fit of the spectral CAPM.** The figure plots model-implied expected excess returns versus realized average excess returns (% per year) for several anomaly portfolios. The model-implied expected returns are obtained using the spectral market component at scale $j = 6$ as the priced factor. Panel (a): We use the first and the last decile portfolio among 24 value-weighted anomaly portfolios as in Hou et al. (2020) (Appendix A.1 contains details). Panel (b): We employ all the anomaly portfolios in Panel (a) with the exception of composite equity issuance (CEI), operating accruals (OA), return on equity and its change (RoE and DRoE) and idiosyncratic and total volatility (IVOL and VAR). The data are monthly from January 1967 through December 2018.

Momentum is a notoriously complicated dimension to price. While it is known that typical cross sections do not load on the momentum factor, a momentum factor is generally required when test portfolios are generated based on momentum (e.g., Fama and French, 2016). Our results in Table 6 support this view. The addition of a momentum factor to the previous cross sections (in Panel B through G) has no material impact on the significance of the business-cycle market factor and on the pricing errors. The use of size and momentum-sorted portfolios (in Panel A) as test assets, however, leads to loadings – as one would expect – on the momentum factor and translates into a misspecification-robust t -statistic for the associated price of risk around 3. In this case, the regression R^2 is a sizable 73% and the regression intercept is significant with a t -statistic around 4. Importantly, adding a business-cycle spectral beta helps. The price of risk associated with the business-cycle market component is 2.78 and, therefore, in line with values estimated for other cross sections. The intercept is now statistically insignificant and the second-stage cross-sectional R^2 is a considerable value of 86%.

We conclude this section with an observation. Our objective is to bring to asset pricing a methodology which is designed to highlight the role of frequency, thereby adding to the more traditional role of style. Consistent with this objective, while the performance of a

Table 6: Cross section of asset prices (with momentum). The table reports the outcome of the cross-sectional regression $\overline{R}_i^e = \lambda_0 + \lambda^{(6)} \widehat{\mathbb{C}}_i^{(6)} + \lambda_{MOM} \widehat{\mathbb{C}}_{i,MOM} + \xi_i$, where $\overline{R}_i^e$ is the mean excess return on portfolio i , $\widehat{\mathbb{C}}_i^{(6)}$ is the corresponding spectral market covariance at scale $j = 6$ and $\widehat{\mathbb{C}}_{i,MOM}$ is the covariance with respect to the momentum factor, MOM_t . In Panel A we use monthly excess returns on the 25 Fama-French portfolios formed on size and momentum. Panels B through G employ the same test assets used in Table 4. The table reports the estimates of the factor risk prices ($\widehat{\lambda}^{(6)}$ and $\widehat{\lambda}_{MOM}$) and the constant term, Fama and MacBeth (1973) standard errors (in parentheses) and the model misspecification-robust standard errors of Kan et al. (2013) (in braces). The last column provides the $\mathbb{R}^2$ of the corresponding cross-sectional regression and its standard error. Estimated values which are significant at the 10% level, or lower, are in bold fonts. We also report the root mean square error (RMSE) and the mean absolute pricing error (MAPE) across all test assets. These are expressed as percentages per year. The data are monthly from January 1967 through December 2018.

Constant	(1) $\lambda^{(6)}$	(2) λ_{MOM}	RMSE	MAPE	$\mathbb{R}^2$
Panel A: 25 Size and Momentum Portfolios					
0.712 (0.227) {0.779}	-0.235 (0.823) {3.913}		3.966	3.143	0.00 (0.04)
0.394 (0.227) {0.490}	2.788 (0.710) {1.305}	0.048 (0.010) {0.021}	1.472	1.261	0.86 (0.11)
0.899 (0.204) {0.2134}		0.039 (0.010) {0.013}	2.077	1.503	0.73 (0.13)
Panel B: 25 Size and Book-to-Market Portfolios					
0.096 (0.266) {0.465}	2.954 (0.818) {1.482}	-0.007 (0.034) {0.070}	1.591	1.078	0.52 (0.22)
Panel C: 25 Size and Profitability Portfolios					
0.235 (0.202) {0.385}	2.957 (0.842) {1.312}	0.030 (0.037) {0.080}	1.122	0.851	0.68 (0.22)
Panel D: 25 Size and Investment Portfolios					
0.245 (0.218) {0.375}	2.696 (1.059) {1.362}	0.010 (0.030) {0.050}	1.708	1.196	0.38 (0.36)
Panel E: 24 Portfolios					
0.588 (0.163) {0.259}	2.124 (0.729) {1.146}	-0.021 (0.027) {0.042}	1.734	1.448	0.46 (0.26)
Panel F: 48 Portfolios from Hou et al. (2020)					
0.568 (0.223) {0.508}	3.669 (0.940) {2.072}	0.014 (0.024) {0.053}	1.387	1.182	0.62 (0.19)
Panel G: 10 Duration Portfolios					
-0.226 (0.582) {2.000}	3.305 (0.858) {1.648}	-0.008 (0.078) {0.258}	1.588	1.240	0.83 (0.14)

single-factor spectral model in which the only factor is a business-cycle market component was shown to be satisfactory along a variety of classical metrics, our intent is not to advocate for a single-factor, business-cycle, market model as a competitor of existing, successful multi-factor models. Rather, our goal is to illustrate the pricing potential of frequency-specific components along with the role that frequency-specific pricing signal may play for dimensionality reduction in the cross section. Searching over frequency, not only over style, can be of aid in bringing order and interpretability to empirical asset pricing while informing modeling work. As in the recent work of, e.g., Feng, Giglio, and Xiu (2020), Freyberger, Neuhierl, and Weber (2020), Harvey et al. (2016), Kelly et al. (2019), Kim, Korajczyk, and Neuhierl (2019) and Kozak et al. (2020), we are interested in more disciplined factor structures. Our proposal is an alternative, albeit complementary to existing recommendations, frequency-based solution.

8. The economics of spectral betas

Using aggregation as a means to assess systematic risk over heterogeneous horizons,²⁵ Kamara et al. (2016) emphasize the importance of delayed adjustments of prices to information in the determination of horizon pricing. It is, therefore, natural to ask what is the link between delayed adjustment models and spectral factor models. In what follows, we show formally that delayed adjustments to information will, in general, lead to frequency-specific spectral betas. Conversely, the heterogeneity of the spectral betas across frequencies will imply (forms of) delayed price adjustments.

We begin with the first implication and show that delayed price adjustments lead, in general, to frequency-specific betas. We work with a model in the spirit of Kamara et al. (2016, pages 1778-1779). Specifically, ignoring (without loss of generality) the intercept, we write

$$y_t = \beta x_t + u_t, \quad (24)$$

where, importantly for our argument, the white noise error term u_t is such that $\mathbb{C}(x_t, u_t) \neq 0$.²⁶ Next, we assume dynamics for the factor x_t in order to “complete” the model specification and highlight the main effects. For illustration, we assume an autoregressive model of order one for the factor and write

$$x_t = \rho x_{t-1} + \varepsilon_t, \quad (25)$$

²⁵For interesting models with heterogeneous investment horizons, see also Beber, Driessen, and Tuijp (2012), Brennan and Zhang (2018) and Crouzet, Dew-Becker, and Nathanson (2020).

²⁶Kamara et al. (2016) correlate u_t and x_{t-1} , rather than u_t and x_t . The change is without loss of generality and is simply meant to cast our discussion in a more natural econometric framework. We will return to the case $\mathbb{C}(x_t, u_{t-1}) \neq 0$ in what follows.

where the white noise error term ε_t is such that $\mathbb{C}(x_t, u_t) = \mathbb{C}(\varepsilon_t, u_t) = \sigma_{\varepsilon, u} \neq 0$. Finally, in light of the non-zero covariance between the residuals, we may write

$$u_t = \frac{\sigma_{\varepsilon, u}}{\sigma_\varepsilon^2} \varepsilon_t + v_t, \quad (26)$$

where v_t is, again, a white noise error term uncorrelated with ε_t . Plugging Eq. (26) into Eq. (24) and, then, Eq. (25) into the resulting expression leads to

$$y_t = \left(\beta + \frac{\sigma_{\varepsilon, u}}{\sigma_\varepsilon^2} \right) x_t - \underbrace{\frac{\sigma_{\varepsilon, u}}{\sigma_\varepsilon^2} \rho x_{t-1}}_{\tilde{v}_t} + v_t, \quad (27)$$

which is explicit about the mechanism through which the information in past (x_{t-1}) and present (x_t) values of the factor are incorporated into asset returns (y_t).

Because $\sigma_{\varepsilon, u} \neq 0$, Eq. (27) is a distributed lag model which, importantly, yields frequency-specific spectral betas. The spectral betas would not be changing with frequency and would, therefore, be equal to the slope β if $\sigma_{\varepsilon, u} = 0$, i.e., precisely in the absence of lagged adjustments to information.

In Figure 5, using the Corollary to Theorem 1 in Appendix A.2, we display the spectral covariances between y_t and x_t , the spectral variances of x_t and the spectral betas between y_t and x_t for the model in Eq. (27). The spectral betas are, of course, ratios between the spectral covariances and the spectral variances. The model is simulated with $\rho = 0.5$, $\sigma_\varepsilon = 1$, $\sigma_u = 1$ and three values of $\sigma_{\varepsilon, u}$, namely $\sigma_{\varepsilon, u} = 0.3$ (in Panel (a)), $\sigma_{\varepsilon, u} = 0$ (in Panel (b)) and $\sigma_{\varepsilon, u} = -0.3$ (in Panel (c)). The three error covariances (positive, zero and negative) yield decreasing, constant and increasing spectral betas, respectively.

The adopted model is purposely simple and illustrative. A richer specification would naturally lead to more complex dynamics in the spectral betas. The key in the three reported cases is, of course, the different impact of the lagged factor x_{t-1} on excess returns y_t . In order to further clarify this statement, we set ρ equal to zero (with $\sigma_{\varepsilon, u} = 0.3$) in Eq. (27) and report the corresponding findings in Panel (d) of Figure 5. Because the partial effect of the lagged factor ($\frac{\sigma_{\varepsilon, u}}{\sigma_\varepsilon^2} \rho$) is now zero, we expect the spectral betas to be constant across frequencies/scales and equal to $\beta + \frac{\sigma_{\varepsilon, u}}{\sigma_\varepsilon^2}$, which is a value of 1.3 given the assumed parameter values. The numbers confirm this logic.²⁷

We now turn to the second implication and show the converse of the previous finding:

²⁷In Panel (d) x_t is white noise. Its spectral density is constant over each interval of frequency $[1/2^j, 1/2^{j-1}]$. Since the spectral variance is the integral of the spectral density over a range of frequency, the variance over the frequencies associated with scale $j = 1, \dots, 10$ is equal to $1/2^j$ and the variance of the process is $\sum_{j=1}^{10} 1/2^j = 1 = \sigma_\varepsilon^2$.

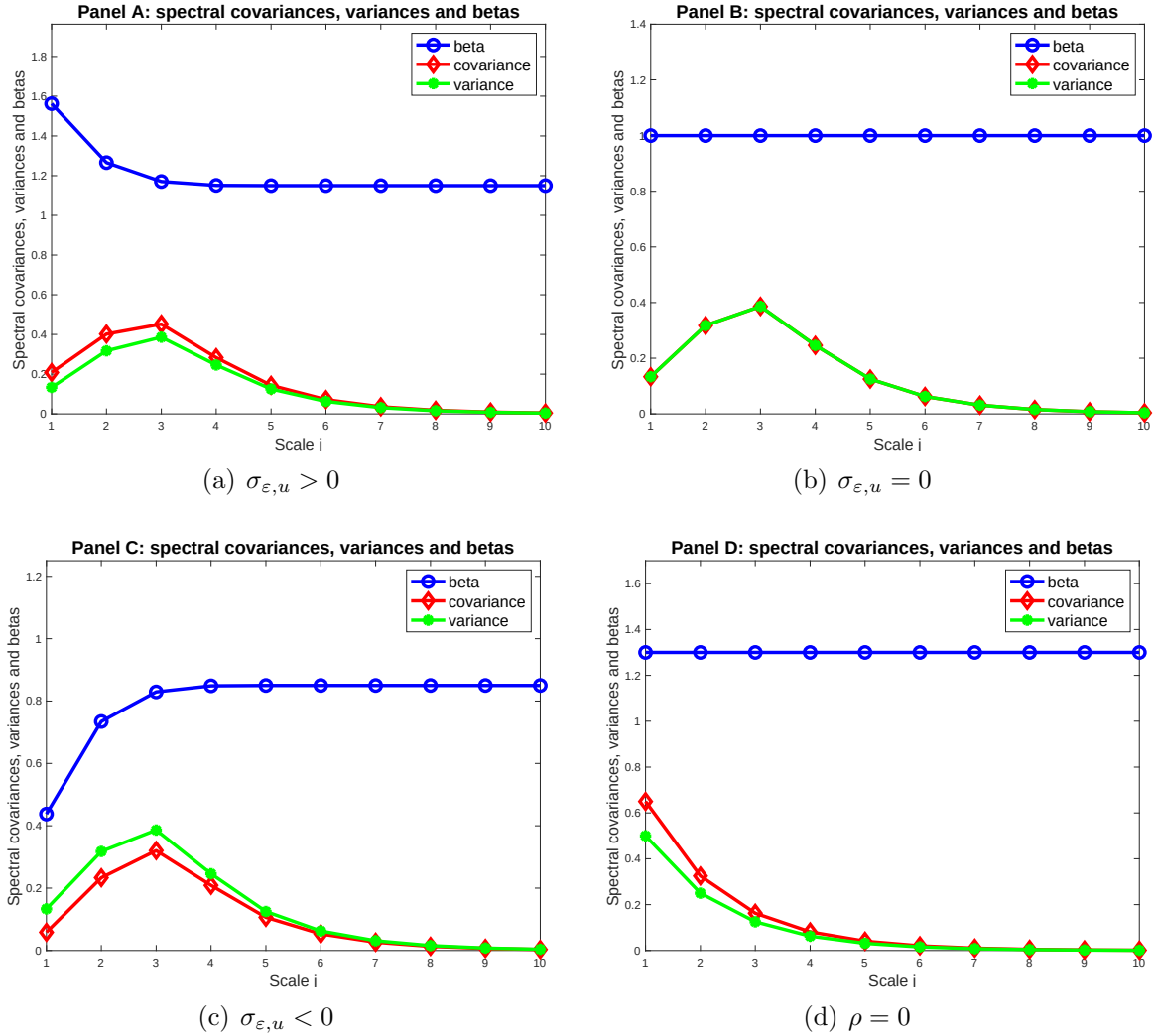


Fig. 5. **Delayed adjustments vs. spectral betas.** In Panels (a), (b), and (c), we display spectral variances, spectral covariances and spectral betas (obtained as in the Corollary to Theorem 1 in Appendix A.2) for three values of the correlation between the factor and the error process. The same quantities are reported in Panel (d) for the case of no dependence in the factor.

frequency-specific betas imply (forms of) delayed price adjustment. To illustrate, we return to the two-beta model in Section 2. For clarity, we set $j^* = 1$ and write

$$y_t = \beta^{HF} x_t^{(1)} + \beta^{LF} x_t^{(>1)} + u_t, \quad (28)$$

where $x_t^{(1)}$ and $x_t^{(>1)} = x_t - x_t^{(1)}$ are two orthogonal components capturing cycles between 1 and 2 periods and larger than 2 periods, respectively. Coherent with the two-beta model in Section 2, the corresponding spectral betas denote high-frequency risk (β^{HF}) and low-frequency risk (β^{LF}).

Now, recalling Eq. (12), we may write

$$x_t = \sum_{k=0}^{\infty} \alpha_k^{2\top} \varepsilon_{t-k},$$

where $\alpha_k^{2\top} = (\alpha_k^3 \ \alpha_k^4)$ and $\varepsilon_t^\top = (\varepsilon_t^1 \ \varepsilon_t^2)$. Because

$$x_t^{(1)} = \left(\frac{\alpha_0^{2\top} - \alpha_1^{2\top}}{\sqrt{2}} \right) \left(\frac{\varepsilon_t - \varepsilon_{t-1}}{\sqrt{2}} \right) + \left(\frac{\alpha_2^{2\top} - \alpha_3^{2\top}}{\sqrt{2}} \right) \left(\frac{\varepsilon_{t-2} - \varepsilon_{t-3}}{\sqrt{2}} \right) + \dots,$$

then

$$\begin{aligned} x_t - x_t^{(1)} &= \left(\frac{\alpha_0^{2\top} + \alpha_1^{2\top}}{\sqrt{2}} \right) \left(\frac{\varepsilon_t + \varepsilon_{t-1}}{\sqrt{2}} \right) + \left(\frac{\alpha_2^{2\top} + \alpha_3^{2\top}}{\sqrt{2}} \right) \left(\frac{\varepsilon_{t-2} + \varepsilon_{t-3}}{\sqrt{2}} \right) + \dots \\ &= \frac{x_{t-1}^{\text{odd}} + x_t^{\text{even}}}{2}, \end{aligned}$$

with

$$\begin{aligned} x_t^{\text{even}} &= (\alpha_0^{2\top} + \alpha_1^{2\top}) \varepsilon_t + (\alpha_2^{2\top} + \alpha_3^{2\top}) \varepsilon_{t-2} + \dots \\ x_{t-1}^{\text{odd}} &= (\alpha_0^{2\top} + \alpha_1^{2\top}) \varepsilon_{t-1} + (\alpha_2^{2\top} + \alpha_3^{2\top}) \varepsilon_{t-3} + \dots \end{aligned}$$

Finally, returning to Eq. (28), we have

$$\begin{aligned} y_t &= \beta^{HF} x_t^{(1)} + \beta^{LF} x_t^{(>1)} + u_t \\ &= \beta^{HF} x_t^{(1)} + \beta^{LF} (x_t - x_t^{(1)}) + u_t \\ &= \beta^{LF} x_t + (\beta^{HF} - \beta^{LF}) x_t^{(1)} + u_t \\ &= \beta^{LF} x_t + (\beta^{HF} - \beta^{LF}) \left(x_t - \left(\frac{x_{t-1}^{\text{odd}} + x_t^{\text{even}}}{2} \right) \right) + u_t \\ &= \beta^{HF} x_t + \underbrace{\left(\frac{\beta^{LF} - \beta^{HF}}{2} \right) (x_{t-1}^{\text{odd}} + x_t^{\text{even}})}_{\tilde{u}_t} + u_t. \end{aligned} \tag{29}$$

As previewed above, our spectral factor model implies lagged effects (given by $x_{t-1}^{\text{odd}} + x_t^{\text{even}}$) and, hence, delayed price/return reactions. Once more, a richer specification than the illustrative (two-beta) specification with $j^* = 1$ considered here would yield richer delayed dependences.

It is interesting to notice that the sign of the correlation between aggregate residuals $\tilde{u}_t$ (comprising of lagged effects) and the aggregate factor x_t depends on the relative magnitude of β^{HF} and β^{LF} . It is negative if $\beta^{HF} > \beta^{LF}$ and positive if $\beta^{HF} < \beta^{LF}$. Because the nature

of the dependence between x_t and $x_{t-1}^{\text{odd}} + x_t^{\text{even}}$ may not be immediate, Figure 6 provides an illustration based on a simulated model in which x_t is, as earlier, an autoregressive process of order one with $\rho = 0.5$, $\sigma_\varepsilon = 1$ and $\sigma_u = 1$. In Panel (a), we set $\beta^{HF} = 1.6$ and $\beta^{LF} = 1.2$ coherently with Panel (a) of Figure 5. In Panel (b), instead, we set $\beta^{HF} = 0.4$ and $\beta^{LF} = 0.8$ coherently with Panel (c) of Figure 5.

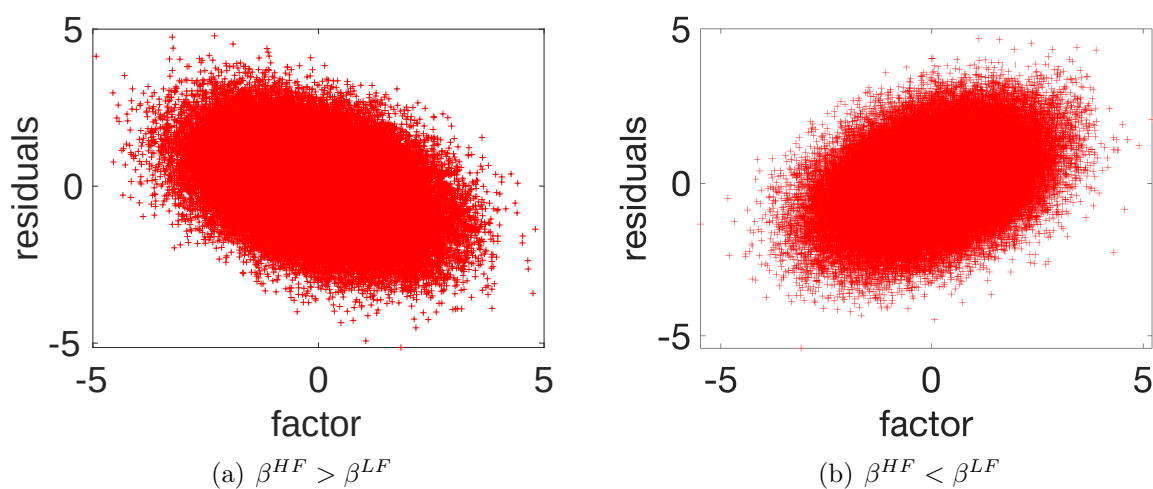


Fig. 6. **Spectral betas vs. delayed adjustments.** Scatter plots between the factor (x_t) and the errors ($\tilde{u}_t$) in Eq. 29 for the case in which the high-frequency spectral beta is larger than the low-frequency spectral beta (Panel (a)) and for the case in which the high-frequency spectral beta is smaller than the low-frequency spectral beta (Panel (b)).

We note that the association between the spectral betas and the sign of the correlation between x_t and aggregate residuals inclusive of lagged effects ($\tilde{v}_t$ in Eq. (27) or $\tilde{u}_t$ in Eq. (29)) is consistent across models. We have commented on the spectral factor model. In the model with lagged adjustments, $\sigma_{\varepsilon,u} > 0$ translates into high-frequency spectral betas larger than their low-frequency counterparts (see Panel (a) of Figure 5). Because $\sigma_{\varepsilon,u} > 0$, however, the factor x_t is negatively correlated with $\tilde{v}_t$. The same logic applies to the case $\sigma_{\varepsilon,u} < 0$.

Importantly, in light of Eq. (27), a regression of y_t on the aggregate factor x_t implies biased and inconsistent estimates of β unless $\sigma_{\varepsilon,u} = 0$, i.e., unless there are no delayed price adjustments or, equivalently, unless systematic risk is stable across frequencies. Similarly, in light of Eq. (29), a regression of y_t on the aggregate factor x_t implies biased and inconsistent estimates of β^{HF} unless $\beta^{HF} = \beta^{LF} = \beta$, i.e., unless systematic risk is stable across frequencies or, equivalently, unless there are no delayed price adjustments.

Next, we study the nature of delayed adjustments for classical characteristics, such as size and value, and link our findings to the structure of the spectral betas across frequencies.

8.1. Size and value

Consider the monthly model

$$R_{p,t} = \alpha_p + \beta_p R_{m,t} + u_{p,t},$$

with $\mathbb{C}(R_{m,t-n}, u_{p,t}) \neq 0$, for $n = 1, \dots, 4$, where $R_{p,t}$ is the return on a generic portfolio p and $R_{m,t}$ is the return on the market index in month t . The model is an extended version of the illustrative specification in Kamara et al. (2016) in which $\mathbb{C}(R_{m,t-1}, u_{p,t}) \neq 0$. Writing $u_{p,t} = \sum_{n=1}^4 \delta_{p,n} R_{m,t-n} + v_{p,t}$, we obtain the estimable econometric specification

$$R_{p,t} = \alpha_p + \beta_p R_{m,t} + \sum_{n=1}^4 \delta_{p,n} R_{m,t-n} + v_{p,t} \quad (30)$$

in which there is no endogeneity, i.e., $\mathbb{C}(R_{m,t-n}, v_{p,t}) = 0$ for $n = 0, \dots, 4$, and the parameters $\delta_{p,n}$ are directly interpretable in terms of delayed (partial) adjustments of the portfolio returns with respect to four previous values of the market return.

Panel (a) and (b) of Figure 7 report the estimated $\delta_{p,n}$ values for quintile portfolios sorted on size along with their t -statistics. Panel (a) and (b) of Figure 8 provide the same information for quintile portfolios sorted on book-to-market. Both in terms of magnitude and in terms of statistical significance, the first delayed adjustment (i.e., the blue bar associated with the one-month lag of the market return) plays a more prominent role than longer

adjustments. It does so for the size-sorted portfolios considerably more than for the value-sorted portfolios. For both sets of portfolios, however, the alignment is monotonic: decreasing for size and increasing for value. Smaller stocks, in particular, are more responsive to past market returns than larger stocks.

In Panel (c) and (d) of Figure 7 we display the first, the second, the third and the sixth spectral betas associated with the five size-sorted portfolios along with their t -statistics. Once more, the same information is contained in Panel (c) and (d) of Figure 8 for the five value-sorted portfolios. We report the first, second and third spectral betas because their relative structure is informative about the role of first-order adjustments, as we discuss below. We report the sixth spectral betas (corresponding to the business-cycle market component) in light of our focus on *priced* systematic risk.

By analyzing Eq. (27), we have previously shown that, when the partial effect associated with lagged price adjustments is positive (i.e., when $\sigma_{\varepsilon,u} < 0$, because of the minus sign in the theoretical slope), the spectral betas increase, albeit at a declining rate (see Figure 5). Conversely, we have shown that when the lagged adjustments have a negative partial effect (i.e., when $\sigma_{\varepsilon,u} > 0$), the spectral betas decrease, again, at a declining rate. These observations are coherent with our findings for size. In fact, the first four portfolios have positive first-order delayed adjustments and increasing (first, second and third) spectral betas whereas the big-firm (fifth) portfolio has a negative first-order delayed adjustment and decreasing spectral betas. Analogous observations can be made about the value portfolios with the caveat that, even though the first-order partial effects are monotonic, they are - both in terms of magnitude and in terms of statistical significance - smaller than for the size portfolios.

We now turn to the sixth spectral betas, namely *priced* risk. In agreement with the corresponding first-order delayed adjustments in Panel (a) of both figures, these betas are monotonically decreasing over size and monotonically increasing over value. Contrary to a first-order delayed adjustment model, however, they appear to be quite different in terms of magnitude from the third spectral beta. As said, a first-order delayed adjustment model would lead to compression of the spectral betas starting with the third beta (c.f., again, Figure 5).

In light of this evidence, we now ask the question: could a delayed adjustment model with one lagged adjustment, as supported by the empirical results in Panels (a) and (b) of Figure 7 and, to a lesser extent, of Figure 8, give rise to the sixth spectral betas in Panel (c)? We address this issue by computing spectral betas from a simulated model with one lagged adjustment, i.e., from Eq. (30) with $\delta_{p,1} \neq 0$ and $\delta_{p,n} = 0$, when $n = 2, 3, 4$. We run 10,000 simulations and simulate 600 monthly observations, as in the data. We find that this model

would find it rather hard to replicate the monotonic pattern in these betas. We count the number of instances across simulations in which we reproduce monotonicity. It is around 25% for size and 4% for value. We also count the number of occurrences so that, for value, the first spectral beta is between 0.7 and 0.8 and the last spectral beta is between 1.3 and 1.4, as in the data. The frequency is around .03%.

We conclude with a few observations. First, the mapping between lead/lag models in the time-domain data generating process and frequency-specific regression slopes in the corresponding frequency-domain linear specifications is an important (albeit not sufficiently emphasized) aspect of regression analysis in the frequency domain. We do not operate in the frequency domain but our emphasis on frequency is such that, unsurprisingly, there should be an important link between delayed partial effects and spectral betas even in our framework. This section was intended to make the link explicit. Second, the focus of the literature on delayed adjustments is on short-term delays. In agreement with that literature, we have found that one-month delays (the longer delays in, e.g., Hou and Moskowitz, 2005) may matter but are more evident in the data, both in terms of magnitude and statistical significance, for size than for value. More generally, their impact should change depending on style. In our framework, these one-month delays contribute to the structure of the spectral betas but do not explain them, particularly at the low frequencies containing pricing signal. Third, longer adjustment (with a likely different economic justification) as well as economic restrictions (as in Subsection 5.1) yield the priced spectral betas. The role of long adjustments is not unique to our framework and has received attention in other areas. Following Campbell and Mankiw (1987), for example, Fuster, Laibson, and Mendel (2010) emphasize that typical model selection criteria, like AIC and BIC, erroneously recommend the use of low-order ARIMA models. High-order models, however, have properties that are not well-captured by low-order models whereas the opposite is not true, the properties of low-order models being rather well-captured by high-order models. In a different context and using alternative (pricing) metrics, our findings provide support for this view. Finally, the pricing implications of delayed adjustment models have generally been analyzed by treating the extent of delayed adjustments as a *characteristic* (c.f. Hou and Moskowitz, 2005), the work on aggregation of Kamara et al. (2016) being an exception. Regardless of the length of the adjustment, spectral factor models may serve as a way to operationalize delayed adjustment models with various levels of complexity, assess their properties across frequencies and derive implications for systematic (frequency-specific) risk.

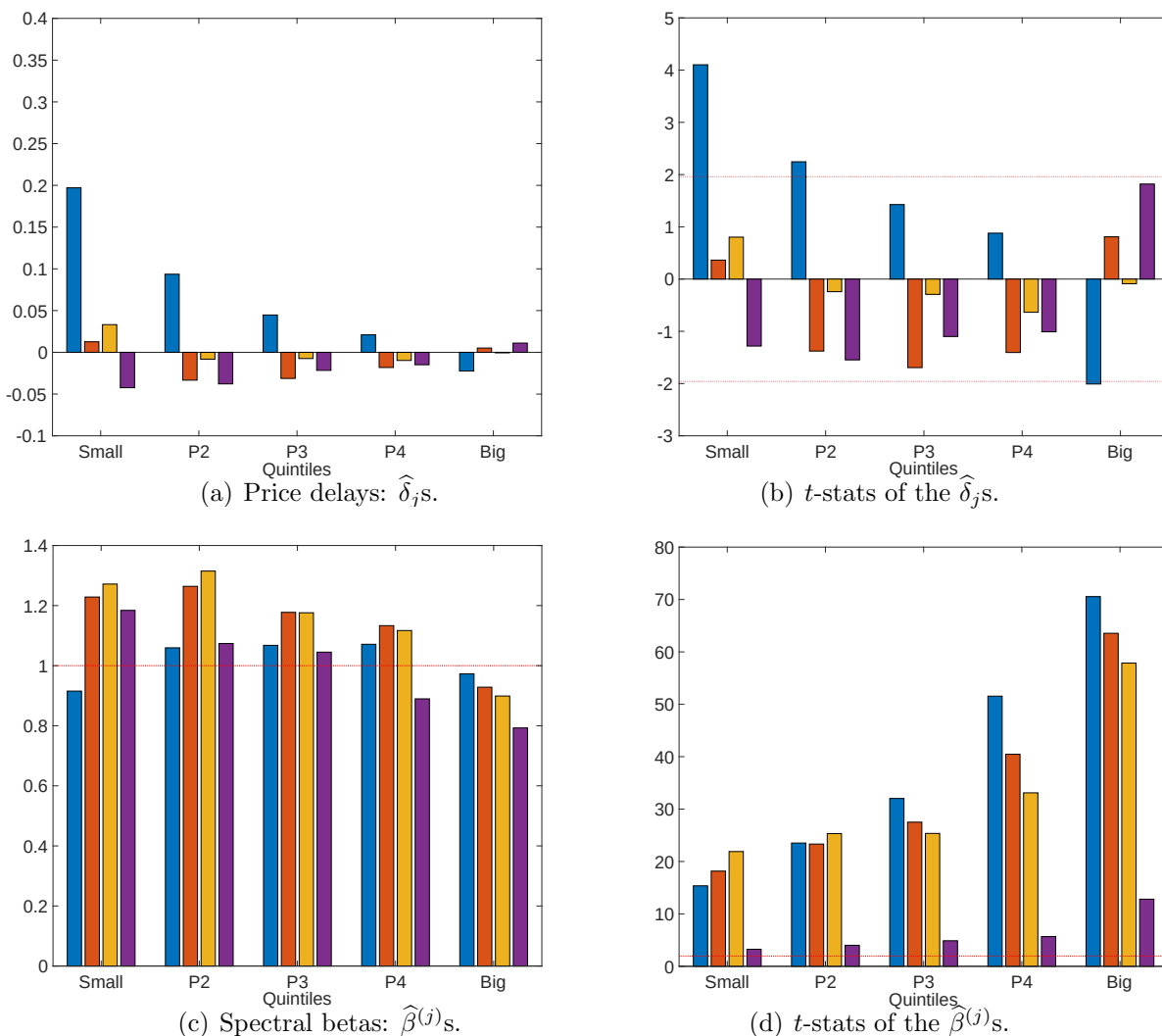


Fig. 7. Size-sorted portfolios: partial effects of four delayed adjustments vs. the spectral betas. Each cluster of bars refers to a specific quintile portfolio, with the smallest quintile denoting “small” firms and the largest quintile denoting “big” firms. The top panels report the estimated δ_j s for $j = 1, \dots, 4$ in Eq. (30) and their Newey-West-adjusted t -statistics. The bottom panels report the estimated spectral betas $\beta^{(j)}$ s, for $j = 1, 2, 3, 6$, where $j = 1, 2, 3, 6$ correspond to economic fluctuations between 1 and 2 months, 2 and 4 months, 4 and 8 months, and 32 and 64 months, respectively. The data are monthly from January 1967 through December 2018.

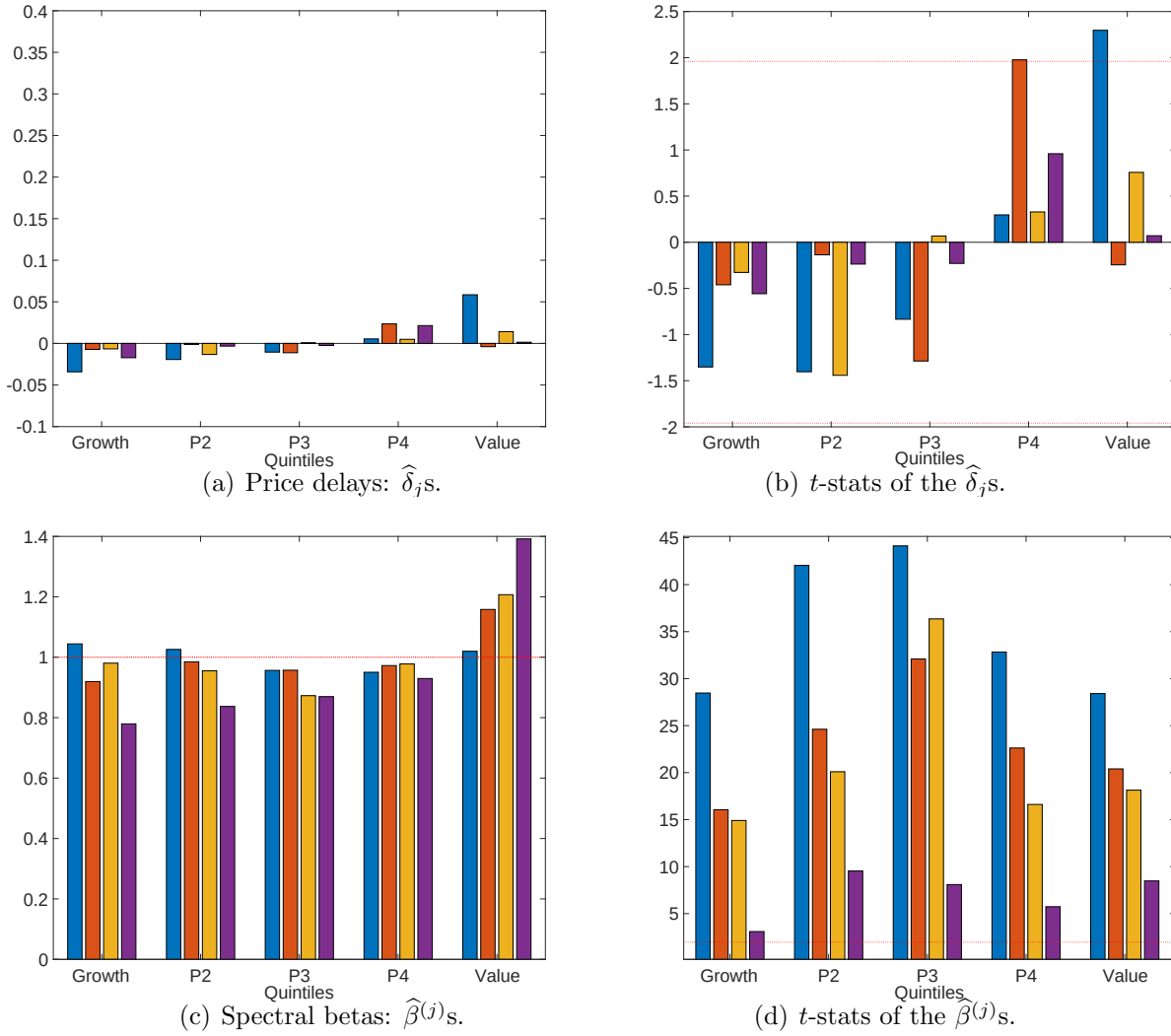


Fig. 8. Value-sorted portfolios: partial effects of four delayed adjustments vs. the spectral betas. Each cluster of bars refers to a specific quintile portfolio, with the smallest quintile denoting “growth” firms and the largest quintile denoting “value” firms. The top panels report the estimated δ_j s for $j = 1, \dots, 4$ in Eq. (30) and the corresponding Newey-West-adjusted t -statistics. The bottom panels report the estimated spectral betas $\beta^{(j)}$ s, for $j = 1, 2, 3, 6$, where $j = 1, 2, 3, 6$ correspond to economic fluctuations between 1 and 2 months, 2 and 4 months, 4 and 8 months, and 32 and 64 months, respectively. The data are monthly from January 1967 through December 2018.

9. Conclusions

Using a suitable Wold representation, we introduce a novel notion of spectral factor model in which the spectral factors are orthogonal components of any factor of interest, from classical economic factors, like the market, to statistical factors, like principal components. The spectral factors represent cycles of the original factor with heterogeneous length. Because of this property, spectral factor models are particularly suited to study the role of frequency in systematic risk assessments. Should systematic risk be constant across frequencies, spectral factor models would coincide with traditional factor models. In this sense, our proposed framework can be viewed as a regression-based generalization of traditional factor models, one in which risk is allowed (without being forced) to vary across frequencies.

The empirical failures of the CAPM are broadly-documented. Yet, the CAPM offers clear implications about the relation between risk and expected returns, thereby providing a natural vantage point for initial investigations into the role of frequency as a means to discipline factor proliferation. Is it believable that noisy high-frequency market dynamics contain limited information about pricing but the pricing signal in slow-moving market components is strong? We show that a single business-cycle component of market returns is effective in pricing an array of interesting cross sections of anomalies. We conclude that attention to frequency, in addition to style, has the potential to lead to more orderly forms of model selection and, ultimately, to theoretically-motivated dimensionality reduction in the factor space.

On the pricing side, future work should study the frequency-specific pricing of factors other than the market, further enlarge the set of test assets and evaluate the potential for combining the proposed methods with alternative dimensionality-reduction techniques. Applications focusing on the portfolio allocation decisions of investors with different preferences, risk appetites and, importantly, investment horizons are, also, interesting avenues for additional explorations.

Appendix A. Appendix

A.1. Data

We obtained the portfolio return data and the factors from three public sources:

1. Kenneth French’s data library available at http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library;
2. AQR’s data library available at <https://www.aqr.com/insights/datasets>
3. Hou, Xue, and Zhang’s data library available at <http://global-q.org/testingportfolios.html>

We use the following set of test assets: (a) 25 value-weighted (VW) portfolios obtained from independent sorts of stocks into size and book-to-market quintiles; (b) 25 VW portfolios obtained from independent sorts of stocks into size and operating profitability quintiles; (c) 25 VW portfolios obtained from independent sorts of stocks into size and investment quintiles; (d) 25 VW portfolios obtained from independent sorts of stocks into size and momentum quintiles; (e) 24 U.S. anomaly portfolios: accruals, betting-against-beta, cash-flow-to-price, investment, earnings-to-price, book-to-market, long-term reversals, net share issuance, quality-minus-junk, profitability, residual variance, market value of equity and short-term reversals. For each anomaly, with the exception of “betting-against-beta” and “quality-minus-junk” for which we use long/short portfolios, we work with the top and the bottom decile portfolios. Except, again, for “betting-against-beta” and “quality-minus-junk”, which are from the AQR’s data library, the return data for all portfolios are from Kenneth French’s library; (f) 48 U.S. anomaly portfolios, taken again as the top and the bottom decile portfolios for each anomaly, from Hou et al. (2020): (Value/growth) book-to-market equity (B/M), cash-flow-to-price (C/P), enterprise multiple (EM), earnings-to-price (E/P), long-term reversal (L-REV) and sales-to-price (S/P); (Investment) operating accruals (OA), composite equity issuance (CEI), discretionary accruals (DAC), net operating assets (NOA), change in net operating assets (dNOA), change in PPE and inventory-to-assets (dPIA), investment-to-assets (I/A), investment growth (IG) and net stock issues (NSI); (Profitability) return on equity (RoE), change in the return on equity (dRoE) and operating profits-to-book equity (Prof/BE); (Intangibles) organizational capital-to-assets (OCA); (Frictions) market equity (ME), market beta (Beta), idiosyncratic volatility (IVOL), short-term reversal (S-REV) and total volatility (VOL); and (g) the 10 duration portfolios of Weber (2018)²⁸.

The factors used in the Fama and French (1993) three-factor model and in the Fama and French (2015) five-factor model are from Kenneth French’s data library. The factors used in

²⁸We thank Michael Weber for kindly making his data available on his website.

Hou et al. (2015) four-factor model are from Hou, Xue, and Zhang's data library.

There are two criteria that guided our choice of anomalies from the universe of 50 used in Hou et al. (2020). First, we wanted to work with a balanced panel. Thus, we removed from the analysis anomalies which start later than 1967 (e.g., portfolios sorted on R&D expense-to-market, which start in July, 1976). Second, because the asymptotic theory implied by the adopted (two-step, misspecification-robust) cross-sectional procedure guarantees reliable inference for $N/T \approx 0.1$, we limited ourselves to the top and the bottom decile portfolios of 24 well-studied anomalies, which delivers a ratio of $N/T \approx 0.08$.²⁹ We, thus, removed from the analysis eight anomalies that Hou et al. (2020) classify as momentum, for which we provide a separate discussion. We also removed five anomalies based on seasonality, which are rarely analyzed in empirical work. Finally, for CEI, DAC, NSI, OA, RoE, dRoE, IVOL, and VOL we compute the high and low return as the average return on the top three and bottom three deciles.

²⁹For observations on the finite sample accuracy of the misspecification-robust methods in Kan et al. (2013), we refer the reader to, e.g., Gospodinov and Robotti (2020).

A.2. Corollary to Theorem 1.

Assume $\mathbf{x} = \{(y_t, x_t)^\top\}_{t \in \mathbb{Z}}$ satisfies Eq. (12). The resulting beta would conform with

$$\beta = \frac{\mathbb{C}[y_t, x_t]}{\mathbb{V}[x_t]} = \sum_{j=1}^{\infty} v^{(j)} \beta^{(j)},$$

where

$$\begin{aligned} \beta^{(j)} = & \frac{\sum_{k=0}^{\infty} \sigma_{\varepsilon^1}^2 \left(\sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+i}^1 - \sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+2^{j-1}+i}^1 \right) \left(\sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+i}^3 - \sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+2^{j-1}+i}^3 \right) / 2^j}{\mathbb{V}[x_t^{(j)}]} \\ & + \frac{\sum_{k=0}^{\infty} \sigma_{\varepsilon^2}^2 \left(\sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+i}^2 - \sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+2^{j-1}+i}^2 \right) \left(\sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+i}^4 - \sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+2^{j-1}+i}^4 \right) / 2^j}{\mathbb{V}[x_t^{(j)}]} \\ & + \frac{\sum_{k=0}^{\infty} \sigma_{\varepsilon^{1,2}} \left(\sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+i}^1 - \sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+2^{j-1}+i}^1 \right) \left(\sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+i}^4 - \sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+2^{j-1}+i}^4 \right) / 2^j}{\mathbb{V}[x_t^{(j)}]} \\ & + \frac{\sum_{k=0}^{\infty} \sigma_{\varepsilon^{1,2}} \left(\sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+i}^2 - \sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+2^{j-1}+i}^2 \right) \left(\sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+i}^3 - \sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+2^{j-1}+i}^3 \right) / 2^j}{\mathbb{V}[x_t^{(j)}]} \end{aligned} \quad (31)$$

with

$$\begin{aligned} \mathbb{V}[x_t^{(j)}] = & \sum_{k=0}^{\infty} \sigma_{\varepsilon^1}^2 \left(\sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+i}^3 - \sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+2^{j-1}+i}^3 \right)^2 / 2^j \\ & + \sum_{k=0}^{\infty} \sigma_{\varepsilon^2}^2 \left(\sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+i}^4 - \sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+2^{j-1}+i}^4 \right)^2 / 2^j \\ & + \sum_{k=0}^{\infty} 2\sigma_{\varepsilon^{1,2}} \left(\sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+i}^3 - \sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+2^{j-1}+i}^3 \right) \left(\sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+i}^4 - \sum_{i=0}^{2^{j-1}-1} \alpha_{k2^j+2^{j-1}+i}^4 \right) / 2^j \end{aligned}$$

and

$$v^{(j)} = \frac{\mathbb{V}[x_t^{(j)}]}{\sum_{k=0}^{\infty} (\sigma_{\varepsilon^1}^2 (\alpha_k^3)^2 + \sigma_{\varepsilon^2}^2 (\alpha_k^4)^2 + 2\sigma_{\varepsilon^{1,2}}^2 \alpha_k^3 \alpha_k^4)}. \quad (32)$$

Proof. Immediate, given the representation in Eq. (13) and its properties. $\square$

A.3. Short-Run Reversal Portfolios

We expand on Section 6 by considering daily, rather than monthly, data. We study another pair of portfolio returns obtained from the market-neutral mean-reversion strategy of Lo (2008). The strategy holds long positions in stocks that underperformed the average stock in the past (specifically, q days ago) and holds short positions in stocks that outperformed the average stock q days ago, i.e., the portfolio weights are $w_{it}(q) = -(R_{i,t-q} - R_{m,t-q})/N$, where $R_{m,t-q} = \sum_i R_{i,t-q}/N$ is the average stock return on date $t-q$ and N is the total number of stocks. The lag q is designed to exploit mean reversion over a specific horizon. We recall that the returns from short-reversal strategies have an interpretation in terms of proxies for the returns to liquidity provision, c.f., Nagel (2012).

Using daily data from July 16, 1962 to December 31, 2016,³⁰ we find that the correlation between the reversal portfolio returns with $q=1$ and $q=2$ is -1.1% . In August 2007, both of these strategies suffered significant losses as part of the “Quant Meltdown.” During that month, the correlation between the $q=1$ and $q=2$ reversal returns spiked to 65.8% . Figure 9 provides a dynamic view of this correlation, computed over 125-day rolling windows, from January 3, 2006 through December 31, 2008. The correlation began increasing in July 2007, but the spike occurred in August 2007. The correlation then declined steadily, until it turned negative in the first half of 2008, only to reverse itself during the second half of the year as the financial crisis began unfolding.

We now turn to frequencies. Because of the use of daily data, Table 1 should be interpreted in terms of daily horizons, rather than in terms of months. In light of the very limited predictability at daily horizons, the identification strategy in Subsection 5.1 is conducted by only including excess market returns in the vector $\tilde{\mathbf{x}}_t$.

First, we note that the two reversal strategies (whose returns are denoted by $R_{1d,t}$ and $R_{2d,t}$, respectively) have similar market betas:

$$\begin{aligned} R_{1d,t} &= \alpha + 0.121 \times R_{m,t} + u_t, \quad \mathbb{R}^2 = 0.02, \\ &\quad (t\text{-stat} = 14.75) \\ R_{2d,t} &= \alpha + 0.123 \times R_{m,t} + u_t, \quad \mathbb{R}^2 = 0.02. \\ &\quad (t\text{-stat} = 16.80) \end{aligned}$$

However, as suggested by the time-varying correlations in Figure 9, the overall market betas

³⁰The strategies are implemented using data from the University of Chicago’s Center for Research in Securities Prices (CRSP). Only U.S. common stocks (CRSP share code 10 and 11) are included, which eliminates REITs, ADRs, and other types of securities. We drop stocks with share prices below \$5 and above \$2,000.

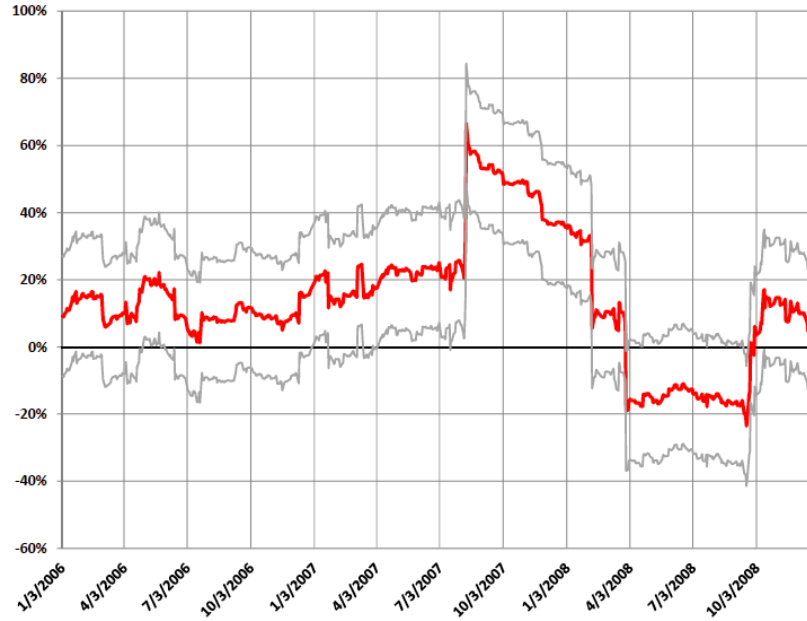


Fig. 9. **Reversal strategy.** We report 125-day rolling-window correlations between daily mean-reversion strategies with allocations $w_{it}(q)$, for $q = 1, 2$ days, where $w_{it}(q) = -(R_{i,t-q} - R_{m,t-q})/N$ and $R_{m,t-q} = \sum_i R_{i,t-q}/N$ is the average stock return on date $t-q$. The gray lines are 2-standard-deviation bands around the estimated correlations. We use daily data from July 16, 1962 to December 31, 2016.

may mask important differences between the two strategies. In effect, the spectral betas behave markedly differently. Table 7 shows the results. The 1-day reversal portfolio has spectral betas which decline considerably and almost monotonically. Not only are the spectral betas drastically attenuated at low frequencies, they even become negative. On the contrary, the spectral betas of the 2-day reversal strategy are mildly hump-shaped, with the first spectral beta being lower than the first spectral beta for the 1-day strategy and the remaining spectral betas being instead larger.

	$j = 1$	$j = 2$	$j = 3$	$j = 4$	$j = 5$	$j = 6$	$j > 6$	$\sum_{j=1}^7 \hat{\mathbb{C}}$
Spectral covariances								
1-day reversal	0.093	0.032	0.006	-0.006	-0.003	-0.003	-0.002	0.117
2-day reversal	0.051	0.041	0.017	0.008	0.003	0.001	-0.000	0.120
	$j = 1$	$j = 2$	$j = 3$	$j = 4$	$j = 5$	$j = 6$	$j > 6$	$\sum_{j=1}^7 \hat{v}^{(j)} \hat{\beta}^{(j)}$
Spectral betas and variance weights								
1-day reversal	0.216	0.113	0.046	-0.098	-0.089	-0.185	-0.126	0.121
weight (rel. variance)	0.444	0.289	0.140	0.064	0.031	0.016	0.016	
2-day reversal	0.118	0.144	0.122	0.129	0.104	0.053	-0.027	0.123
weight (rel. variance)	0.443	0.290	0.141	0.063	0.031	0.016	0.016	

Table 7: Short-run reversal portfolios: covariance and beta decomposition. The panels provide spectral covariances, spectral betas and relative variance weights associated with seven frequencies for two short-run reversal portfolios. We decompose the portfolios (excess) returns and the market (excess) returns into six orthogonal components in addition to a residual term $\pi_t^{(6)}$, as discussed in Section 5. All returns are in percent (i.e., the raw covariances are multiplied by 10^4).

Table 8 shows the results when we bundle together frequencies below (about) three weeks and above (about) three weeks.³¹ Again, we find substantial differences between the two strategies at both high and low frequencies. At high frequencies, the betas of the 1-day reversal strategy are larger than those associated with the 2-day reversal strategy. At low frequencies, the betas of the 2-day reversal strategy are positive but insignificant, whereas those of the 1-day reversal strategy are negative and significant. This behavior is hidden when working with an aggregate market beta.

Once again, consistently with our theoretical argument regarding the orthogonality of the components, the multiple regressions and the simple regressions deliver similar estimates.

	Simple regression		Multiple regression	
	β^{LF}	β^{HF}	β^{LF}	β^{HF}
1-day reversal	-0.120 (-3.811)	0.137 (5.374)	-0.153 (-2.760)	0.140 (5.199)
2-day reversal	0.058 (1.309)	0.129 (6.990)	0.009 (1.176)	0.131 (6.845)

Table 8: Short-run reversal portfolios. We run simple and multiple regressions on high-frequency and low-frequency market components for two short-run reversal portfolios. We decompose the portfolios (excess) returns and the market (excess) returns into six orthogonal components in addition to a residual term $\pi_t^{(6)}$, as discussed in Section 5. We obtain the high-frequency market component by summing the market components up to scale $j = 4$ (included). The sum of the remaining market components defines the low-frequency market component. The table provides estimated betas and t -statistics.

³¹We use a cut-off $j = 4$ which, with daily data, corresponds to $2^4/5 = 3.2$ weeks.

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